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Galactic Chemical Evolution Models Favor an Extended Type Ia Supernova Delay-Time Distribution

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ABSTRACT

Type Ia supernovae (SNe Ia) produce most of the Fe-peak elements in the Universe and therefore are a crucial ingredient in galactic chemical evolution models. SNe Ia do not explode immediately after star formation and the delay-time distribution (DTD) has not been definitively determined by supernova surveys or theoretical models. Because the DTD also affects the relationship among age, [Fe/H], and [α/Fe] in chemical evolution models, comparison with observations of stars in the Milky Way is an important consistency check for any proposed DTD. We implement several popular forms of the DTD in combination with multiple star formation histories for the Milky Way in one-zone and multi-zone chemical evolution models using the Versatile Integrator for Chemical Evolution (VICE). VICE includes a prescription for radial stellar migration in multi-zone models. We compare our predicted interstellar medium abundance tracks, stellar abundance distributions, and stellar age distributions to the final data release of the Apache Point Observatory Galactic Evolution Experiment (APOGEE). We find that the SN Ia DTD has the largest effect on the [α/Fe] distribution: a DTD with more prompt SNe Ia produces a stellar abundance distribution that is skewed toward a lower [α/Fe] ratio, whereas a DTD with more delayed SNe Ia produces a distribution that is skewed toward higher [α/Fe]. While the DTD alone cannot explain the observed bimodality in the [α/Fe] distribution, in combination with an appropriate star formation history it affects the goodness of fit between the simulated and observed high-α sequence. Our model results favor a DTD with longer delay times and fewer prompt SNe Ia than the fiducial t^{-1} power law.

1. INTRODUCTION

Galactic chemical evolution (GCE) studies seek to explain the observed distribution of metals throughout the Milky Way Galaxy. Tinsley (1979) made a compelling case that the non-Solar [α/Fe][1] ratios seen by, e.g., Wallerstein (1962) were caused by different stellar lifetimes for the contributors of the Fe-peak elements than the α-elements. Type Ia supernovae (SNe Ia), the thermonuclear explosions of carbon-oxygen white dwarfs (WDs), are responsible for a majority of the Fe produced in the Galaxy (Matteucci & Greggio 1986); meanwhile, core collapse supernovae (CCSNe), the explosions of massive stars, produce the α-elements (e.g., O and Mg) in addition to a smaller fraction of Fe. SNe Ia are delayed by $\sim 10^8 - 10^{10}$ years after star formation events, as evidenced by observations at large distances from star forming regions and in elliptical galaxies (e.g., Maza & van den Bergh 1976). This delayed enrichment leads to a decrease in [α/Fe] with increasing [Fe/H] (Matteucci & Greggio 1986). Therefore, the relative abundances of the α-elements and Fe as a function of stellar age trace the balance of SN rates over time.

The rate of SNe Ia which explode as a function of time after the formation of a hypothetical unit-mass single stellar population is known as the delay-time distribution (DTD; for a review, see Section 3.5 of Maoz et al. 2014). When the DTD is convolved with the Galactic star formation rate (SFR) as a function of time, it yields the SN Ia rate as a function of time. The quantitative details of the relationship between [α/Fe] and [Fe/H] are set by the DTD, and as such it is a key parameter in GCE models. The DTD remains poorly constrained, however, because it reflects the detailed evolution of the SNe Ia progenitor systems, so different models for the progenitors of SNe Ia will naturally predict different forms for the DTD.

The mechanism(s) for producing SNe Ia are not fully understood (for reviews, see Maoz et al. 2014; Livio &

[1] In standard bracket notation, $[X/Y] \equiv \log_{10}(X/Y) - \log_{10}(X/Y)_\odot$. In this work we will use [α/Fe] and [O/Fe] interchangeably, although observational studies will often use a combination of α-elements to calculate [α/Fe].

Mazzali 2018; Ruiter 2020). Two general production channels have been proposed. In the single-degenerate (SD) case, the WD accretes mass from a close non-degenerate companion until it surpasses ~ 1.4 $M_\odot$ and explodes (Whelan & Iben 1973; Nomoto 1982; Yoon & Langer 2003). In the double-degenerate (DD) case, two WDs merge after a gravitational-wave inspiral (Iben & Tutukov 1984; Webbink 1984; Pakmor et al. 2012) or head-on collision (Benz et al. 1989; Thompson 2011). Searches for signs of interaction between the SN ejecta and a non-degenerate companion (e.g., Panagia et al. 2006; Chomiuk et al. 2016; Fausnaugh et al. 2019; Tucker et al. 2020; Dubay et al. 2022) or for a surviving companion (e.g., Schaefer & Pagnotta 2012; Do et al. 2021; Tucker & Shappee 2023) have placed tight constraints on the SD channel, heavily disfavoring it as the main pathway for producing "normal" SNe Ia. The DD channel, now the preferred model, faces issues of its own with matching observed SN Ia rates because not all WD mergers may lead to a thermonuclear explosion (e.g., Nomoto & Iben 1985; Saio & Nomoto 1998; Shen et al. 2012), and the progenitor systems are difficult to detect even within our own Galaxy (Rebassa-Mansergas et al. 2019).

Uncertainties in the progenitor channels of SNe Ia translate into uncertainties in the shape of the DTD. For the DD channel, assumptions about the distribution of post-common envelope separations and the rate of gravitational wave inspiral suggest a broad $\sim t^{-1}$ DTD at long delay times ($\gtrsim 1$ Gyr), but at short delays ($\lesssim 1$ Gyr) the rate is limited by the need to produce two WDs (see Section 3.5 from Maoz et al. 2014). Higher-order progenitor systems could also produce a t^{-1} DTD (Fang et al. 2018; Rajamuthukumar et al. 2023). The DTD which would result from the SD channel depends greatly on the assumptions of binary population synthesis, but in general is expected to cover a narrower range of delay times and may feature a steep exponential cutoff at the long end (e.g., Greggio 2005).

Surveys of SNe Ia have provided the primary observational constraints on the DTD. This can be done by comparing the observed rate of SNe Ia to their host galaxy parameters (e.g., Mannucci et al. 2005; Heringer et al. 2019) or inferred star formation histories (SFHs; e.g., Maoz et al. 2012), measuring SN Ia rates in galaxy clusters (e.g., Maoz et al. 2010), or comparing the volumetric SN Ia rate to the cosmic SFH as a whole (e.g., Graur et al. 2014; Strolger et al. 2020). Early studies, which had limited sample sizes, produced unimodal (Strolger et al. 2004) or bimodal (Mannucci et al. 2006) DTDs where the majority of SNe Ia explode within a relatively narrow range of delay times. More recent studies have recovered broader DTD functions, with many converging on a declining power-law of $\sim t^{-1}$ (e.g., Maoz & Graur 2017; Castrillo et al. 2021; Wiseman et al. 2021), though there is some evidence for a steeper slope in galaxy clusters (Maoz & Graur 2017; Friedmann & Maoz 2018). It is especially difficult to constrain the DTD for short delay times (Maoz & Mannucci 2012) because of the need for SN Ia rates at long lookback times and uncertainties in the age estimates of stellar populations.

The uncertainties in the SN Ia DTD propagate into GCE models. In principle, then, the observed chemical abundance patterns should contain information about the DTD, and therefore the progenitors, of SNe Ia. A few studies have compared different DTDs in one-zone chemical evolution models, but their comparisons to abundance data are limited to the local Solar neighborhood (e.g., Andrews et al. 2017; Palicio et al. 2023). Poulhazan et al. (2018) compared six DTD models in a smoothed-particle hydrodynamics simulation in a cosmological context. They found that models with a prompt SN Ia component produced the highest abundance ratios and the narrowest [α/Fe] distributions at late times, although their simulation was not tuned to reproduce the parameters of the Milky Way and they did not compare to observed chemical abundances. The current era of large spectroscopic surveys such as the Apache Point Observatory Galactic Evolution Experiment (APOGEE; Majewski et al. 2017) and the forthcoming Milky Way Mapper [cite] has made abundances across the Milky Way disk available for comparison to more sophisticated GCE models.

This paper presents the first comprehensive look at the DTD in a multi-zone GCE model that can qualitatively reproduce the observed abundance structure of the Milky Way disk. A multi-zone approach allows for a radially-dependent parameterization of the SFH, outflows, stellar migration, and abundance gradient which can better match observations across the Galactic disk. We evaluate a selection of DTD models from the literature with multiple SFHs and a prescription for radial stellar migration in the Versatile Integrator for Chemical Evolution (VICE; Johnson & Weinberg 2020). In Section 2, we present our models for the DTD and SFH, and we describe our chemical evolution models. In Section 3, we present the results of our one-zone chemical evolution models. In Section 4, we present the results of our multi-zone models and compare to observations. In Section 5, we discuss the implications for the DTD and future surveys. In Section 6, we summarize our conclusions.

2. METHODS

We make use of the multi-zone simulation tools in VICE developed by Johnson et al. (2021) (hereafter J21). The basic setup of our models follows theirs: the simulated Galaxy is a disk with a radius of 20 kpc and is divided into concentric rings of width $\delta R_{\rm gal} = 100$ pc. The gas in each ring is assumed to be instantaneously mixed; while star particles are allowed to migrate radially (see Appendix C), there is no exchange of gas between rings. We run our simulations with a time-step size of $\Delta t = 10$ Myr up to a maximum time of $t_{\rm max} = 13.2$ Gyr. Following J21, we set VICE to form $n = 8$ stellar populations per ring per time-step, and we set a maximum star-formation radius of $R_{\rm SF} = 15.5$ kpc, such that $\dot{\Sigma}_* = 0$ for $R_{\rm gal} > R_{\rm SF}$ (stars are allowed to migrate past $R_{\rm SF}$, they are just not born there). We summarize the parameters for our multi-zone models in Table 1.

2.1. *Nucleosynthetic Yields*

Like J21, we focus our analysis on O and Fe, representing the α and Fe-peak elements, respectively. Both elements are produced by CCSNe. VICE adopts the instantaneous recycling approximation for CCSNe, so the equation which governs CCSN enrichment as a function of star formation for some element x is simply

$$\dot{M}_x^{\rm CC} = y_x^{\rm CC}\dot{M}_* \tag{1}$$

where $y_x^{\rm CC}$ is the CCSN yield of element x per unit mass of star formation, and $\dot{M}_*$ is the SFR. Following J21, who in turn adopt their CCSN yields from Chieffi & Limongi (2004) and Limongi & Chieffi (2006), we adopt $y_{\rm O}^{\rm CC} = 0.015$ and $y_{\rm Fe}^{\rm CC} = 0.0012$. The primary effect of these yields is to set the low-[Fe/H] "plateau" in [O/Fe] which represents pure CCSN enrichment. The chosen yields for this paper produce a plateau at [O/Fe] = 0.45; see Weinberg et al. (2023) for more discussion on the effect of the CCSN yields on chemical evolution.

Following the formalism of Weinberg et al. (2017), the rate of Fe contribution to the ISM from SNe Ia is

$$\dot{M}_{\rm Fe}^{\rm Ia} = y_{\rm Fe}^{\rm Ia}\langle\dot{M}_*\rangle_{\rm Ia} \tag{2}$$

where $\langle\dot{M}_*\rangle_{\rm Ia}$ is the time-averaged SFR weighted by the DTD and $y_{\rm Fe}^{\rm Ia}$ is the Fe yield of SNe Ia. Weinberg et al. (2017) show in their Appendix A that

$$\langle\dot{M}_*\rangle_{\rm Ia} \equiv \frac{\int_0^t \dot{M}_*(t')R_{\rm Ia}(t-t')dt'}{\int_{t_D}^{t_{\rm max}} R_{\rm Ia}(t')dt'} \tag{3}$$

where $R_{\rm Ia}$ is the DTD in units of $\rm M_\odot^{-1}\,yr^{-1}$, t_D is the minimum SN Ia delay time, and $t_{\rm max}$ is the maximum simulation time. The denominator of Equation 3 is therefore the total number of SNe Ia per $\rm M_\odot$ of stars formed.

The yield $y_{\rm Fe}^{\rm Ia}$ measures the mass of Fe produced by SNe Ia per unit mass of star formation. It can be expressed in terms of the integral of the DTD:

$$y_{\rm Fe}^{\rm Ia} = m_{\rm Fe}^{\rm Ia}\int_{t_D}^{t_{\rm max}} R_{\rm Ia}(t')dt' = m_{\rm Fe}^{\rm Ia}\frac{N_{\rm Ia}}{M_*}, \tag{4}$$

where $m_{\rm Fe}^{\rm Ia}$ is the average mass of Fe produced by a single SN Ia, and $N_{\rm Ia}/M_* = 2.2\times10^{-3}\,\rm M_\odot^{-1}$ is the average number of SNe Ia per mass of stars formed (Maoz & Mannucci 2012). Adjusting the value of $y_{\rm Fe}^{\rm Ia}$ primarily affects the end point of chemical evolution tracks in [O/Fe]–[Fe/H] space. Following J21, we adopt $y_{\rm Fe}^{\rm Ia} = 0.00214$. This yield was originally adapted from the W70 model of Iwamoto et al. (1999), but was slightly modified so that the inside-out SFH model produces stars with [O/Fe] ≈ 0.0 by the end of the simulation. Palla (2021) studies the effect of different SN Ia yields on GCE models in detail.

2.2. *DTD Models*

We explore five different functional forms for the SN Ia DTD: a power-law, a broken power-law with an initial flat plateau, an exponential, a two-population model, and a model based on triple-system dynamics, with a one or two variants of each. Figure 1 presents these DTDs as a function of time after star formation, and Table 2 summarizes the DTD models and their parameters. We use simple forms rather than simulated physical or analytical models of SNe Ia for the sake of decreased computational time. Additionally, physically-motivated models of the DTD must contend with many unknown or poorly-constrained parameters, so our simple forms allow us to reduce the number of free parameters. In Appendix B, we show that a few of our simple forms adequately approximate the more complete analytical models of Greggio (2005).

In this subsection we present unnormalized functional forms $f_{\rm Ia}$ for each DTD model. The relationship between $f_{\rm Ia}$, which has units of $\rm Gyr^{-1}$, and $R_{\rm Ia}$, which has units of $\rm M_\odot^{-1}\,Gyr^{-1}$, is:

$$R_{\rm Ia}(t) = \begin{cases} \frac{N_{\rm Ia}}{M_*}\frac{f_{\rm Ia}(t)}{\int_{t_D}^{t_{\rm max}} f_{\rm Ia}(t')dt'}, & t \geq t_D \\ 0 & t < t_D. \end{cases} \tag{5}$$

VICE automatically normalizes the provided function for the DTD according to this equation.

Table 1. A summary of parameters and their fiducial values for our chemical evolution models. We omit parameters that are unchanged from J21; see their Table 1 for details.

Quantity	Fiducial Value(s)	Section	Description
$R_{\rm gal}$	[0, 20] kpc	2	Galactocentric radius
$\delta R_{\rm gal}$	100 pc	2	Width of each concentric ring
$\Delta R_{\rm gal}$	N/A	C	Change in orbital radius due to stellar migration
$p(\Delta R_{\rm gal}\vert\tau, R_{\rm form})$	Equation C1	C	Probability density function of radial migration
z	[-3, 3] kpc	C	Distance from Galactic midplane at present day
$p(z\vert\tau, R_{\rm final})$	Equation C2	C	Probability density function of Galactic midplane distance
Δt	10 Myr	2	Time-step size
$t_{\rm max}$	13.2 Gyr	2	Maximum simulation time
n	8	2	Number of stellar populations formed per ring per time-step
$R_{\rm SF}$	15.5 kpc	2	Maximum radius of star formation
$M_{g,0}$	0	2.3	Initial gas mass
$R_{\rm Ia}(t)$	Equation 5	2.2	Delay time distribution of Type Ia supernovae
t_D	40 Myr	2.2	Minimum SN Ia delay time
$N_{\rm Ia}/M_*$	2.2×10^{-3} $\rm M_\odot^{-1}$	2.1	SNe Ia per unit mass of stars formed (Maoz & Mannucci 2012)
$y_{\rm O}^{\rm CC}$	0.015	2.1	CCSN yield of O
$y_{\rm Fe}^{\rm CC}$	0.0012	2.1	CCSN yield of Fe
$y_{\rm O}^{\rm Ia}$	0	2.1	SN Ia yield of O
$y_{\rm Fe}^{\rm Ia}$	0.00214	2.1	SN Ia yield of Fe
$f_{\rm IO}(t\vert R_{\rm gal})$	Equation 11	2.3	Time-dependence of the inside-out SFR
$f_{\rm LB}(t\vert R_{\rm gal})$	Equation 12	2.3	Time-dependence of the late-burst SFR
$\tau_{\rm EB}(t)$	Equation 13	2.3	Time-dependence of the early-burst SFE timescale
$f_{\rm EB}(t\vert R_{\rm gal})$	Equation 14	2.3	Time-dependence of the early-burst IFR
$f_{\rm TI}(t\vert R_{\rm gal})$	Equation 16	2.3	Time-dependence of the two-infall IFR

Table 2. Summary of SN Ia DTD models explored in this paper.

Model	Eq.	Parameters	Similar to
Power-law	6	$\alpha = -1.1$	Maoz & Graur (2017, field); Wiseman et al. (2021)
Power-law	6	$\alpha = -1.4$	Maoz & Graur (2017, cluster); Heringer et al. (2019)
Plateau	7	$W = 0.3$ Gyr, $\alpha = -1.1$	Greggio (2005, CLOSE DD)
Plateau	7	$W = 1.0$ Gyr, $\alpha = -1.1$	Greggio (2005, WIDE DD)
Triple-system	8	$f_{\rm init} = 0.05 f_{\rm peak}$, $t_{\rm rise} = 0.5$ Gyr, $W = 0.5$ Gyr, $\alpha = -1.1$	Rajamuthukumar et al. (2023)
Exponential	9	$\tau = 1.5$ Gyr	Greggio (2005, SD); Schönrich & Binney (2009); Weinberg et al. (2017)
Exponential	9	$\tau = 3.0$ Gyr	—
Two-population	10	$t_{\rm max} = 0.05$ Gyr, $\sigma = 0.015$ Gyr, $\tau = 3.0$ Gyr	Mannucci et al. (2006)

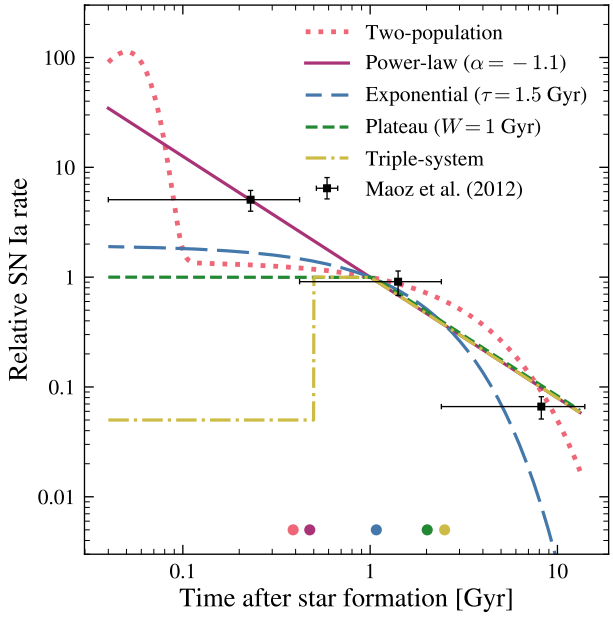

Figure 1. Selection of models for the SN Ia delay time distribution (DTD) used in this paper. All functions are normalized such that $f_{\rm Ia}(t = 1\,{\rm Gyr}) = 1$. The black squares represent the DTD recovered for the SDSS-II sample of SNe Ia by Maoz et al. (2012) at the same scale as the DTD models. The horizontal and vertical error bars indicate the time range and 1σ uncertainties, respectively, of each DTD measurement. The colored circles along the horizontal axis indicate the median delay time for each model.

Power-law—A simple power law with slope α:

$$f_{\rm Ia}^{\rm plaw}(t) = (t/1\,{\rm Gyr})^{\alpha} \tag{6}$$

A declining power-law with $\alpha \sim -1$ is expected to arise from typical assumptions about the distribution of post-common envelope separations and the rate of gravitational wave inspiral (see Section 3.5 from Maoz et al. 2014). It is therefore a commonly assumed DTD in GCE studies (e.g., Rybizki et al. 2017; J21). Additionally, the observational evidence for a power-law DTD is strong. Maoz & Graur (2017) obtain a DTD with $\alpha = -1.07 \pm 0.09$ based on volumetric rates and an assumed cosmic SFH for field galaxies in redshift range $0 \leq z \leq 2.25$. Wiseman et al. (2021) obtain a similar slope of $\alpha = -1.13 \pm 0.05$ for field galaxies in the redshift range $0.2 < z < 0.6$. For galaxy clusters, Maoz & Graur (2017) find a steeper DTD slope of $\alpha = -1.39^{+0.32}_{-0.05}$. Heringer et al. (2019) use a SFH-independent method to constrain the DTD for field galaxies within $0.01 < z < 0.2$ and find a steeper slope of $\alpha = -1.34^{+0.19}_{-0.17}$. In this paper, we investigate the cases $\alpha = -1.1$ and $\alpha = -1.4$.

Plateau—A modification of the simple power-law in which the DTD "plateaus" for a duration W before declining:

$$f_{\rm Ia}^{\rm plat}(t) = \begin{cases} 1, & t < W \\ (t/W)^{\alpha}, & t \geq W \end{cases} \tag{7}$$

Our primary motivation is to consider a model which matches observations at delay times beyond a few Gyr, where the DTD is best constrained, but which has a much smaller fraction of prompt ($\lesssim 100$ Myr) SNe Ia than the simple power law. To our knowledge, this form of the DTD has not been considered in previous studies. However, we show in Appendix B that this form can approximate the more complicated analytical DD DTDs from Greggio (2005). We investigate the cases $W = 0.3$ Gyr and $W = 1$ Gyr, taking $\alpha = -1.1$ for all plateau models.

Triple-system—A DTD based on simulations of triple-system evolution by Rajamuthukumar et al. (2023). We approximate their numerically-generated DTD as a special case of the plateau model (Equation 7) where the initial rate is quite low until an instantaneous rise to the plateau value at time $t_{\rm rise}$:

$$f_{\rm Ia}^{\rm triple}(t) = \begin{cases} \epsilon, & t < t_{\rm rise} \\ 1, & t_{\rm rise} \leq t < W \\ (t/W)^{\alpha}, & t \geq W \end{cases} \tag{8}$$

with $t_{\rm rise} = 0.5$ Gyr, $W = 0.5$ Gyr, $\alpha = -1.1$, and $\epsilon = 0.05$ so the initial rate is 5% of the peak rate.

Exponential—An exponentially declining DTD with timescale τ:

$$f_{\rm Ia}^{\rm exp}(t) = \frac{1}{\tau} e^{-t/\tau} \tag{9}$$

This model can be used to calculate an analytic solution to the equations of chemical equilibrium, making it a popular choice among GCE studies. Schönrich & Binney (2009) and Weinberg et al. (2017) both assume an exponential DTD with a timescale $\tau = 1.5$ Gyr. We show in Appendix B that a $\tau = 1.5$ Gyr exponential is an adequate approximation for the analytical SD DTD from Greggio (2005). There is less observational support for an exponential DTD. Strolger et al. (2020), fitting to the cosmic SFH and SFHs from field galaxies, find a range of exponential-like solutions with timescales $\sim 1.5 - 6$ Gyr. In this paper we investigate timescales $\tau = 1.5$ and 3 Gyr.

Two-population—A DTD in which $\sim 50\%$ of SNe Ia belong to a "prompt" Gaussian component at small t and the remainder form an exponential tail at large t:

$$f_{\rm Ia}^{\rm twopop}(t) = 0.5\left[e^{-\frac{(t-t_p)^2}{2\sigma^2}} + e^{-t/\tau}\right] \tag{10}$$

To approximate the DTD from Mannucci et al. (2006), we take $t_p = 50$ Myr, $\sigma = 15$ Myr, and $\tau = 3$ Gyr, which results in $\sim 50\%$ of SNe Ia exploding within $t < 100$ Myr. This formulation is slightly different than the approximation used in other GCE studies (e.g., Matteucci et al. 2006; Poulhazan et al. 2018), where it has a more distinctly bimodal shape. We have compared the two approximations to this DTD in a one-zone model and found that they produce very similar abundance tracks and MDFs. This DTD is adopted by the Feedback In Realistic Environments (FIRE; Hopkins et al. 2014) and FIRE-2 (Hopkins et al. 2018) simulations.

2.2.1. *The Minimum SN Ia Delay Time*

In addition to the DTD shape, the minimum SN Ia delay time t_D is another parameter which can have an effect on chemical evolution observables, such as the location of the high-α knee and the [O/Fe] distribution function (DF) (Andrews et al. 2017). The value of t_D is set by the lifetime of the most massive SN Ia progenitor system. Previous GCE studies have adopted values ranging from $t_D \approx 30$ Myr (e.g., Poulhazan et al. 2018) to $t_D = 150$ Myr (e.g., J21). We take $t_D = 40$ Myr to be our fiducial value as it is the approximate lifetime of an $8\,\mathrm{M}_\odot$ star. In Section 3 we find that adopting a longer t_D has only a minor effect on the chemical evolution for most DTD models except the power-law, but in that case the effect of a longer t_D can be approximated by adding an initial plateau of width $W = 0.3$ Gyr to the DTD (see Figure 4).

2.3. *Star Formation Histories*

We consider four models for the SFH: inside-out, late-burst, early-burst, and two-infall. The former two models, which feature a smooth SFH, were investigated by J21 using a similar methodology to this work. The inside-out model produced a better agreement to the age–[O/Fe] relation observed by Feuillet et al. (2019), while the late-burst model better matched their observed age–metallicity relation. The latter two models (early-burst and two-infall) feature discontinuous or "bursty" SFHs, and were proposed by Conroy et al. (2022) and Chiappini et al. (1997), respectively, to explain the distinct evolution of the high- and low-α sequences. Together, these four models cover a range of behavior including a smooth SFH and SFR-, SFE-, and infall-driven starbursts.

The inside-out and late-burst models are run in VICE's "star formation mode," where the SFR surface density $\dot{\Sigma}_*$ is prescribed along with the star formation efficiency (SFE) timescale $\tau_* \equiv \Sigma_g/\dot{\Sigma}_*$. The remaining quantities, infall rate (IFR) surface density $\dot{\Sigma}_{\rm in}$ and gas surface density Σ_g, are calculated from the specified quantities. The latter two models are run in "infall mode," where we specify $\dot{\Sigma}_{\rm in}$ and τ_*. The mode in which VICE models are run makes no difference as a unique solution can always be obtained if two of the four equations are specified. See Appendix B of J21 for details on the normalization of the SFH. All models have zero initial gas mass in all zones. We present an overview of these models in Figure 2.

Inside-out—As in J21, this is our fiducial SFH. The dimensionless time-dependence of the SFR is given by

$$f_{\rm IO}(t|R_{\rm gal}) = (1 - e^{-t/\tau_{\rm rise}})e^{-t/\tau_{\rm sfh}}, \tag{11}$$

where we assume $\tau_{\rm rise} = 2$ Gyr for all radii. The SFH timescale $\tau_{\rm sfh}$ varies with $R_{\rm gal}$, with $\tau_{\rm sfh}(R_{\rm gal} = 8\,\mathrm{kpc}) \approx 15$ Gyr at the Solar annulus and longer timescales in the outer Galaxy. The $\tau_{\rm sfh} - R_{\rm gal}$ relation is determined from the data of Sánchez (2020); see Section 2.5 of J21 for the details.

Late-burst—A variation on the inside-out SFH with a Gaussian burst in the SFR at late times:

$$f_{\rm LB}(t|R_{\rm gal}) = f_{\rm IO}(t|R_{\rm gal})\Big(1 + A_b e^{-(t-t_b)^2/2\sigma_b^2}\Big), \tag{12}$$

where A_b is the dimensionless amplitude of the starburst, t_b is the time of the peak of the burst, and σ_b is the width of the Gaussian. Evidence for a recent star formation burst $\sim 2-3$ Gyr ago has been found in *Gaia* (Mor et al. 2019) and in massive WDs in the Solar neighborhood (Isern 2019). Following J21, we adopt $A_b = 1.5$, $t_b = 11.2$ Gyr, and $\sigma_b = 1$ Gyr. The determination of $\tau_{\rm sfh}$ is the same as in the inside-out case.

Early-burst—An extension of the model proposed by Conroy et al. (2022) to explain the non-monotonic behavior of the high-α sequence down to [Fe/H] ≈ -2.5. An abrupt factor ~ 20 rise in the SFE at early times drives an increase in the [O/Fe] abundance and populates the high-α sequence. We adopt the formula for the time-dependence of the SFE timescale from Conroy et al. (2022):

$$\frac{\tau_{\rm EB}}{1\,\mathrm{Gyr}} = \begin{cases} 50, & t < 2.5\,\mathrm{Gyr} \\ \frac{50}{[1+3(t-2.5)]^2}, & 2.5 \le t \le 3.7\,\mathrm{Gyr} \\ 2.36, & t > 3.7\,\mathrm{Gyr}. \end{cases} \tag{13}$$

In contrast to the previous SFH models, we run the early-burst model in VICE's "infall" mode. While Conroy et al. (2022) use a constant infall rate in their one-zone model, we adopt a radially-dependent infall rate which declines exponentially with time:

$$f_{\rm EB}(t|R_{\rm gal}) = e^{-t/\tau_{\rm sfh}}, \tag{14}$$

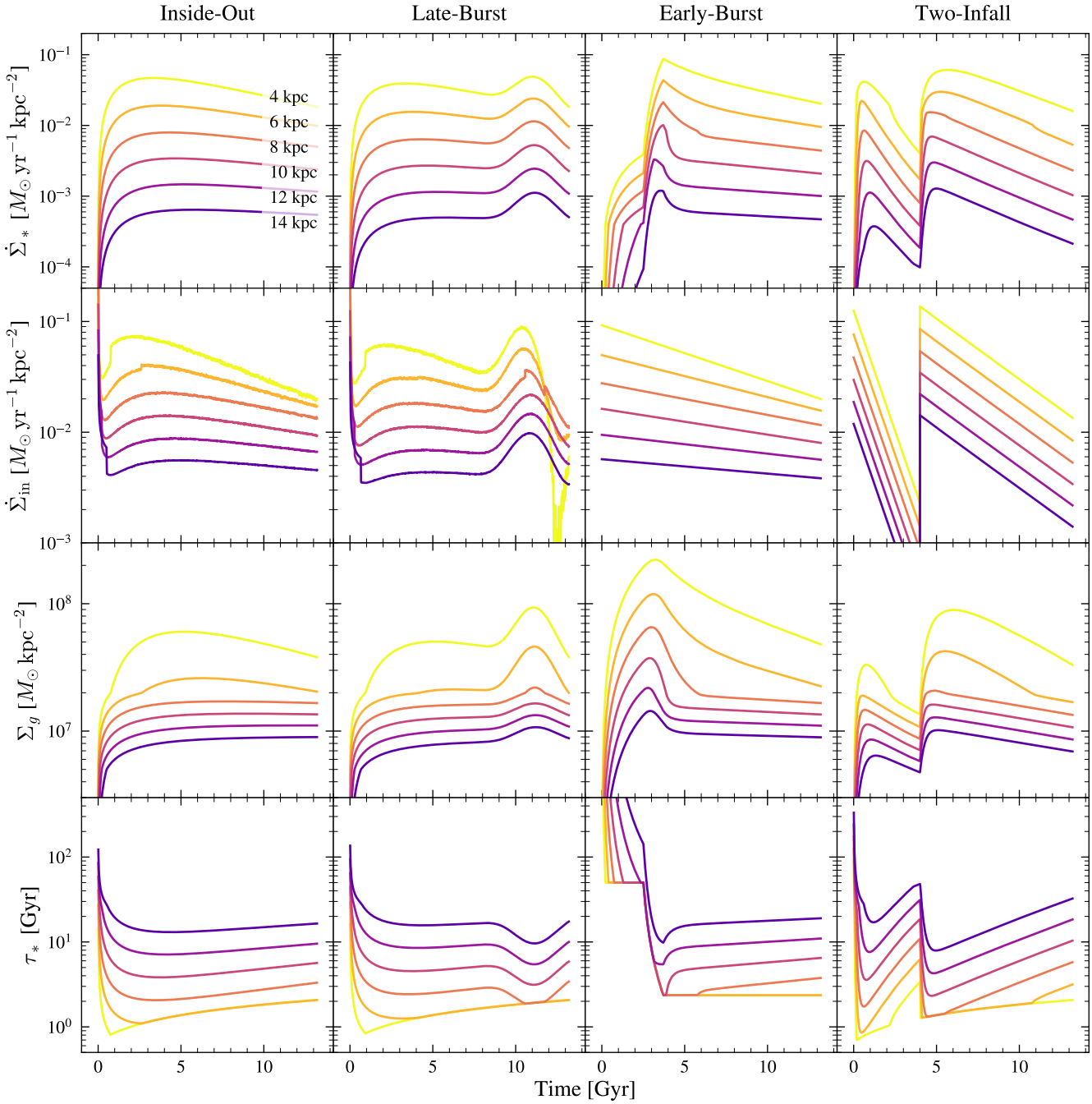

Figure 2. The surface densities of star formation $\dot{\Sigma}_*$ (first row), gas infall $\dot{\Sigma}_{\rm in}$ (second row), and gas mass Σ_g (third row), and the SFE timescale τ_* (fourth row) as functions of simulation time for our four model SFHs: inside-out (first column; see Equation 11), late-burst (second column; see Equation 12), early-burst (third column; see Equations 14 and 13), and two-infall (fourth column; see Equation 16). In each panel, we plot curves for the simulation zones which have inner radii at 4 kpc (yellow), 6 kpc (orange), 8 kpc (red), 10 kpc (violet), 12 kpc (indigo), and 14 kpc (blue).

where $\tau_{\rm sfh}$ is the same as in the inside-out case. To calculate $\dot{\Sigma}_*$ from the above quantities, we modify the fiducial star formation law adopted from J21, substituting $\tau_{\rm EB}$ for the SFE timescale of molecular gas:

$$\dot{\Sigma}_* = \begin{cases} \Sigma_g \tau_{\rm EB}^{-1}, & \Sigma_g \geq \Sigma_{g,2} \\ \Sigma_g \tau_{\rm EB}^{-1} \left(\frac{\Sigma_g}{\Sigma_{g,2}}\right)^{2.6}, & \Sigma_{g,1} \leq \Sigma_g \leq \Sigma_{g,2} \\ \Sigma_g \tau_{\rm EB}^{-1} \left(\frac{\Sigma_{g,1}}{\Sigma_{g,2}}\right)^{2.6} \left(\frac{\Sigma_g}{\Sigma_{g,1}}\right)^{0.7}, & \Sigma_g \leq \Sigma_{g,1}, \end{cases} \tag{15}$$

with $\Sigma_{g,1} = 5 \times 10^6\,{\rm M}_\odot\,{\rm kpc}^{-2}$ and $\Sigma_{g,2} = 2 \times 10^7\,{\rm M}_\odot\,{\rm kpc}^{-2}$.

Two-infall—First proposed by Chiappini et al. (1997), this model parameterizes the infall rate as two successive, exponentially declining bursts to explain the origin of the high- and low-α disk populations:

$$f_{\rm TI}(t|R_{\rm gal}) = N_1(R_{\rm gal})e^{-t/\tau_1} + N_2(R_{\rm gal})e^{-(t-t_{\rm on})/\tau_2}, \tag{16}$$

where N_1 and N_2 are the normalizations of the first and second infall, respectively, and $t_{\rm on}$ is the onset time of the second infall. The normalization ratio N_2/N_1 is calculated so that the thick-to-thin-disk surface density ratio $f_\Sigma(R) = \Sigma_2(R)/\Sigma_1(R)$ is given by

$$f_\Sigma(R) = f_\Sigma(0)e^{R(1/R_2 - 1/R_1)}. \tag{17}$$

Following Bland-Hawthorn & Gerhard (2016), we adopt values for the thick disk scale radius $R_1 = 2.0$ kpc, thin disk scale radius $R_2 = 2.5$ kpc, and $f_\Sigma(0) = 0.27$. We note that some previous studies which use the two-infall model (e.g., Spitoni et al. 2021) do not consider gas outflows, and instead adjust the nucleosynthetic yields to reproduce the Solar abundance. We adopt radially-dependent outflows as in J21 (see their Section 2.4 for details) for all our SFH models, including two-infall. We discuss the implications of this difference in Section 5.2.

2.4. *Observational Sample*

We compare our simulation results to abundance measurements from the final data release (DR17; Abdurro'uf et al. 2022) of the Apache Point Observatory Galactic Evolution Experiment (APOGEE; Majewski et al. 2017). APOGEE used infrared spectrographs (Wilson et al. 2019) mounted on two telescopes: the 2.5-meter Sloan Foundation Telescope (Gunn et al. 2006) at Apache Point Observatory, United States in the Northern Hemisphere, and the Irénée DuPont Telescope (Bowen & Vaughan 1973) at Las Campanas Observatory, Chile in the Southern Hemisphere. After the spectra were passed through the data reduction pipeline (Nidever et al. 2015), the APOGEE Stellar Parameter and Chemical Abundance Pipeline (ASPCAP; Holtzman et al. 2015; García Pérez et al. 2016) extracted chemical abundances using the model grids and interpolation method described by Jönsson et al. (2020).

We restrict our sample to red giant branch and red clump stars with high-quality spectra. Table 3 lists our selection criteria, which largely follow from Hayden et al. (2015). This produces a final sample of 192 990stars with [O/Fe] and [Fe/H] abundance measurements. APOGEE stars are matched with their Bailer-Jones photo-geometric distance estimate from *Gaia* Early Data Release 3 (Gaia Collaboration et al. 2016, 2021), which we use to calculate galactocentric radius $R_{\rm gal}$ and midplane distance z. We use estimated ages from Leung et al. (2023) (hereafter L23), who use a variational encoder-decoder network to retrieve age estimates for APOGEE giants without contamination from age-abundance correlations. Importantly, the L23 sample does not experience the ~ 10 Gyr plateau that affects ages from astroNN (Mackereth et al. 2019). We use an age uncertainty cut of 40% per the recommendations of L23, which produces a total sample of 58 987APOGEE stars with age estimates. We note that we use the full sample of 192 990APOGEE stars unless we explicitly compare to age estimates. Table 4 presents the median uncertainties in the data for [Fe/H], [O/Fe], and age.

3. RESULTS FROM ONE-ZONE MODELS

Before running the full multi-zone simulations, it will be useful to observe the effects of the DTD in more idealized conditions. A one-zone model assumes the entire gas reservoir is instantaneously mixed, so there is no spatial dependence for the distribution of metal abundances. This limits the ability to compare to observations across the disk, but it allows the models to be run in a fraction of the time and eliminates the complicating factor of stellar migration. In this section, we compare the results from one-zone models which examine various parameters of the DTD while keeping other parameters fixed.

All models are run for a duration of $t_{\rm max} = 13.2$ Gyr with a time-step of $\Delta t = 0.01$ Gyr. We adopt the inside-out SFR (Equation 11) evaluated at $R_{\rm gal} = 8$ kpc (i.e., $\tau_{\rm rise} = 2$ Gyr and $\tau_{\rm sfh} = 15.1$ Gyr), an SFE timescale $\tau_* \equiv M_g/\dot{M}_* = 2$ Gyr, and an initial gas mass $M_{g,0} = 0$. We adopt a continuous recycling prescription (see Equation 2 from Johnson & Weinberg 2020). Unless otherwise specified, we adopt an outflow mass-loading

Table 3. Sample selection parameters from APOGEE DR17

Parameter	Range	Notes
$\log g$	$1.0 < \log g < 3.8$	Select giants only
$T_{\rm eff}$	$3500 < T_{\rm eff} < 5500$ K	Reliable temperature range
S/N	$S/N > 80$	Required for accurate stellar parameters
ASPCAPFLAG Bits	$\notin 23$	Remove stars flagged as bad
EXTRATARG Bits	$\notin$ 0, 1, 2, 3, or 4	Select main red star sample only
Age	$\sigma_{\rm Age} < 40\%$	Age uncertainty from L23

Table 4. Median Parameter Uncertainties

Parameter	Median Uncertainty	Source
[Fe/H]	9.2×10^{-3}	APOGEE DR17
[O/Fe]	1.8×10^{-2}	APOGEE DR17
log(Age/Gyr)	0.10	L23

Table 5. A summary of parameters and their fiducial values for our one-zone chemical evolution models.

Quantity	Value	Description
Δt	10 Myr	Time-step size
$t_{\rm max}$	13.2 Gyr	Maximum simulation time
η	2.15	Outflow mass-loading factor
τ_*	2.0 Gyr	SFE timescale
$\dot{M}_r$	continuous	Recycling rate
t_D	40 Myr	Minimum SN Ia delay time
$M_{g,0}$	0	Initial gas mass
$\dot{M}_*(t)$	Equation 11	Star formation rate
$\tau_{\rm rise}$	2.0 Gyr	SFR rise timescale
$\tau_{\rm sfh}$	15.1 Gyr	SFR exponential timescale

factor $\eta \equiv \dot{M}_{\rm out}/\dot{M}_* = 2.15$ (see Equation 8 from J21) and a minimum SN Ia delay time of 40 Myr. A summary of these fiducial parameters is presented in Table 5.

The left-hand panel of Figure 3 compares the results of three one-zone models which are identical except for the slope of the power-law DTD. A steeper slope produces a sharper "knee" and a faster decline in [O/Fe], resulting in a narrower distribution of [O/Fe] around the low-α sequence and a dearth of high-α stars. Conversely, a shallower slope produces a gentler knee and slower decline in [O/Fe], resulting in a broader [O/Fe] DF and a greater number of high-α stars. In all cases the [O/Fe] DF is distinctly unimodal. The [Fe/H] DF is not as strongly affected by the power-law slope: a shallower slope results in only a modest increase in the width of the distribution.

Similar trends can be seen when adjusting the timescale of the exponential DTD, as shown in the middle panel of Figure 3. Here, the knee is not a sharp feature associated with the onset of SNe Ia as in the power-law case, but rather a gentle curve in the abundance track around a simulation time of 1 Gyr. A short exponential timescale shifts this knee down to lower [O/Fe] values, while a longer timescale raises it. The effect on the [O/Fe] DF is similar to the power-law slope: a short timescale produces more low-α stars, while a long timescale produces more high-α stars (but still a unimodal distribution). The effect on the [Fe/H] DF is somewhat more pronounced, with longer timescales skewing to lower [Fe/H] values. A doubling of the timescale from 1.5 Gyr to 3 Gyr raises the [O/Fe] abundance ratio at $t = 1$ Gyr by ~ 0.05 dex and at $t = 3$ Gyr by ~ 0.1 dex, but the equilibrium abundance reached at $t = 10$ Gyr remains unchanged. For all exponential models, the low-α peak sits approximately 0.02 dex lower than for the power-law models.

Finally, the right-hand panel of Figure 3 shows the effect of the plateau DTD model width on abundance tracks and distributions. The effect on the [O/Fe] DF is similar to the previous two models: a longer plateau raises the median delay time, resulting in a smaller low-α peak and a more prominent high-α tail. The abundance tracks from several different plateau widths fill the space in between the exponential ($\tau = 3$ Gyr) and power-law ($\alpha = -1.1$ with no plateau) models, which are both included in the panel for reference. The abundance track from the plateau ($W = 1$ Gyr) model is nearly identical to the exponential ($\tau = 3$ Gyr), although the [O/Fe] DFs are somewhat more distinct, with the plateau model producing a peak closer to the power-law case.

One parameter of the DTD which we hold fixed in the multi-zone models is the minimum SN Ia delay time t_D.

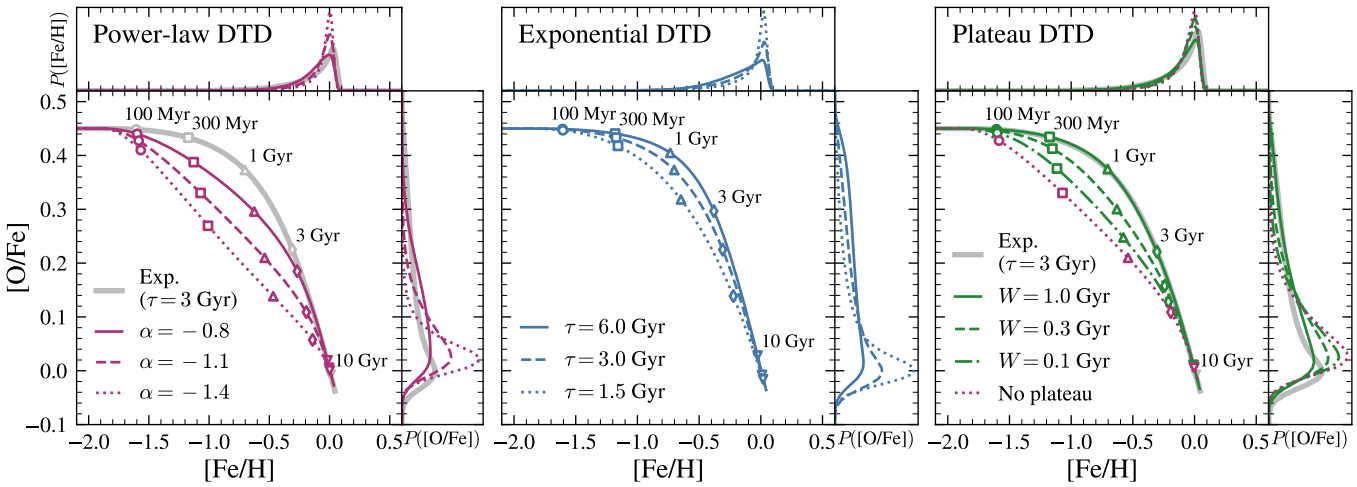

Figure 3. *Left:* Abundance tracks in the [Fe/H]–[O/Fe] plane for one-zone chemical evolution models models which assume a power-law DTD. The dotted, dashed, and solid curves represent models with power-law slopes $\alpha = -0.8$, -1.1, and -1.4, respectively. For reference, the solid gray curve represents an exponential DTD with $\tau = 3$ Gyr. In all cases a minimum SN Ia delay time of 40 Myr is assumed. The open symbols located along each curve mark logarithmic steps in simulation time. The marginal panels above and to the right of the central panel present the distribution functions (DFs) of [Fe/H] and [O/Fe], respectively. For display purposes, these distributions are convolved with a Gaussian kernel with a standard deviation of 0.02 dex. *Middle:* Similar, comparing exponential DTDs with varying timescales. The solid, dashed, and dotted curves represent timescales $\tau = 6$, 3, and 1.5 Gyr, respectively. *Right:* Similar, comparing plateau DTDs with varying plateau widths. The solid, dashed, and dotted green curves represent widths $W = 1$, 0.3, and 0.1 Gyr, respectively. All plateau DTD models assume a post-plateau power-law slope of $\alpha = -1.1$. For reference, the solid gray curve represents an exponential DTD with $\tau = 3$ Gyr, and the dotted purple curve represents a power-law DTD with $\alpha = -1.1$ and no plateau.

Typical choices for t_D in the literature range from ~ 30 Myr (e.g., Poulhazan et al. 2018) to 150 Myr (e.g., J21). The left-hand panel of Figure 4 shows that t_D has a much stronger effect on the abundance tracks and distributions of models which assume a power-law DTD than models which assume a plateau or exponential DTD. This is a consequence of the much higher number of prompt SNe Ia in the power-law models compared to the others (see Figure 1). Moreover, a power-law DTD with a long t_D may be observationally hard to distinguish from a plateau model. In Figure 4, the abundance track for the model with a power-law DTD and $t_D = 150$ Myr (dashed purple line) is similar to that of the plateau DTD with $W = 0.3$ Gyr and $t_D = 40$ Myr (solid green line), and their [O/Fe] DFs are virtually identical. The difference between the two values of t_D for the exponential ($\tau = 3$ Gyr) DTD in both the abundance tracks and distributions is nearly indistinguishable. We do not consider the effect on the other DTD models because a 150 Myr minimum delay time is incompatible with the two-population model, which has $\sim 50\%$ of SNe Ia explode in the first 100 Myr, and would have a negligible effect on the triple-system DTD due to its low SN Ia rate at short delay times.

The right-hand panel of Figure 4 compares the one-zone model outputs from the full range of DTD models we investigate in this work. As with the individual DTD parameters, the choice of DTD primarily affects the location of the high-α "knee" in the [Fe/H]–[O/Fe] abundance tracks. At one extreme is the triple-system model, which sees the CC SN plateau extend up to [Fe/H] ≈ 0.8 followed by a sharp downward turn as the SN Ia rate suddenly increases at a delay time of 500 Myr. Star formation proceeds for such a long time before the knee that the [O/Fe] DF shows a slight second peak around the CC SN value; this model is the only one which produces any amount of bimodality in the one-zone simulations. At the other extreme are the two-population and power-law ($\alpha = -1.1$) models which peak immediately after the minimum delay time of 40 Myr, placing the knee at [Fe/H] ≈ -1.8. The two-population model has a unique second knee at [Fe/H] ≈ -0.2 and [O/Fe] ≈ 0.1, which is a result of the delayed exponential component. The abundance tracks from the plateau ($W = 1$ Gyr) and exponential ($\tau = 1.5$ Gyr) models lie in the intermediate space between these extremes. On the other hand, the [O/Fe] DF from the exponential model has the strongest peak, while the plateau ($W = 1$ Gyr) and triple-system models produce nearly identical distributions in the low-α regime. The power-law ($\alpha = -1.1$) and two-population DTDs produce similar [O/Fe] DFs despite notably different abundance tracks. The exponential ($\tau = 3$ Gyr) and plateau ($W = 0.3$ Gyr) models, while not shown, produce similar abundance tracks to

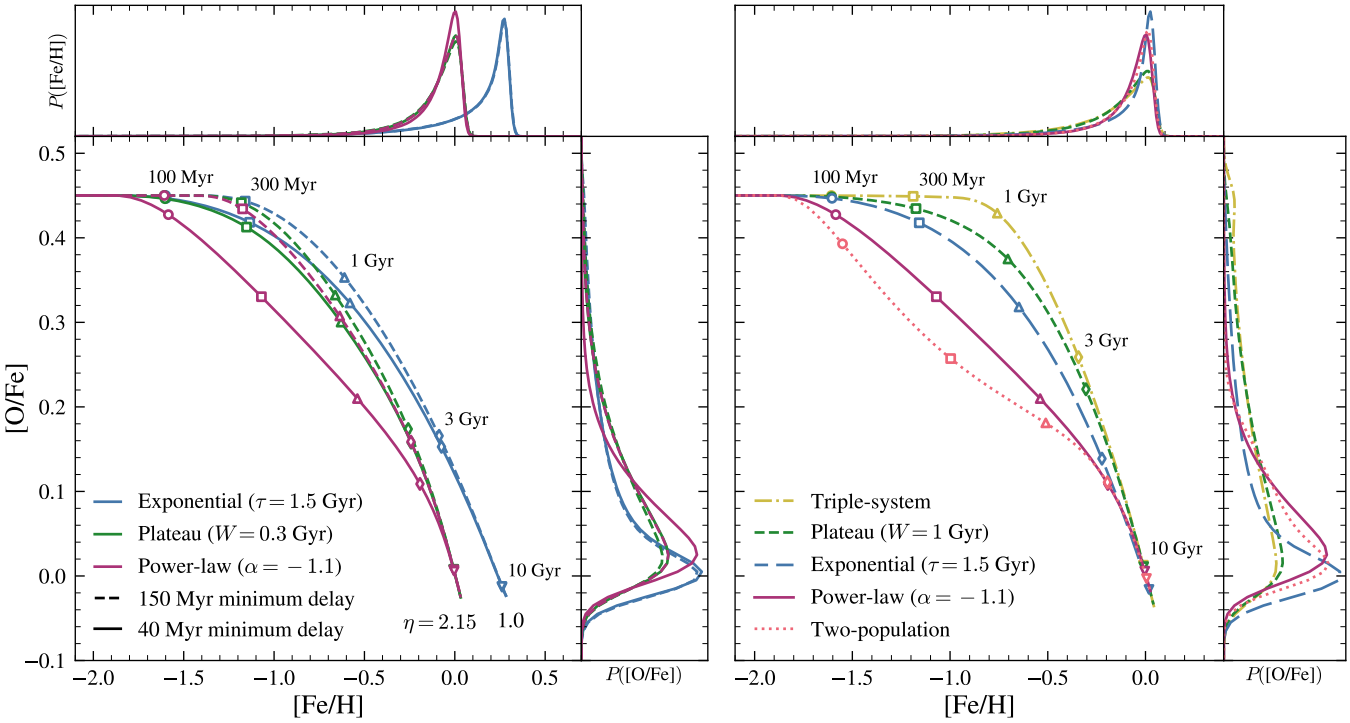

Figure 4. *Left:* Comparison of one-zone models with minimum SN Ia delay times of 40 Myr (solid curves) and 150 Myr (dashed curves) for two DTDs: a power-law with $\alpha = -1.1$ (blue curves), and an exponential with $\tau = 1.5$ Gyr (purple curves). The layout is similar to the plots in Figure 3. For clarity, we assume a mass-loading factor $\eta = 1$ for the exponential DTD curves. *Right:* Comparison of one-zone models which assume all five functional forms for the DTD in this paper: triple-system (green), plateau with $W = 1$ Gyr (cyan), exponential with $\tau = 1.5$ Gyr (purple), power-law with $\alpha = -1.1$ (blue), and two-population (pink). A minimum SN Ia delay time of 40 Myr is assumed for all models in the right panel.

the plateau ($W = 1$ Gyr) and exponential ($\tau = 1.5$ Gyr) models, respectively.

The results presented in this section indicate that the observables which should show the greatest effect from the choice of DTD are the [O/Fe]–[Fe/H] abundance tracks and the [O/Fe] DF. Degeneracies between models in one regime, such as the abundance tracks from the exponential ($\tau = 3$ Gyr) and plateau ($W = 1$ Gyr) DTDs, can be resolved in the other. Of course, both of these observables are also greatly affected by the parameters of the SFH. In this section we focused on the fiducial inside-out SFH model; for a similar comparison of DTD models with a two-infall SFH, we direct the reader to Palicio et al. (2023).

4. RESULTS FROM MULTI-ZONE MODELS

4.1. *The distribution of [Fe/H]*

Figure 5 shows MDFs across the Galaxy for a selection of models and APOGEE data. The two left-hand columns show MDFs from our simulations which assume the same DTD (exponential with $\tau = 1.5$ Gyr) but different model SFHs. For comparison, the APOGEE MDFs are binned similarly and presented in the center column of Figure 5.

Holding the SFH fixed, varying the DTD has a very minor effect on the MDF across the disk. The two right-hand columns of Figure 5 plot the MDFs for two multi-zone simulations which both assume an inside-out SFH but different DTDs: a power-law with slope $\alpha = -1.4$, and an exponential with timescale $\tau = 3$ Gyr. The balance between prompt and delayed SNe Ia is starkly different between the two models, with $\sim 80\%$ of SNe Ia exploding within 1 Gyr in the former model but only $\sim 30\%$ in the latter.

The choice of DTD and SFH has a larger effect on the inner regions of the galaxy than the outer. Inner regions have a higher SFR and more stars are formed at early times than late, so differences in the SN Ia rate have a more pronounced effect on the MDF. For all SFH models except two-infall, the outer galaxy has a longer SFR timescale than the inner galaxy, so the MDF in the outer galaxy is relatively unaffected by the fraction of prompt SNe Ia.

While there are some quantitative differences in how the shape of the MDF varies with Galactic region, the qualitative trends are unaffected by the choice of model SFH or DTD. These trends are primarily driven by chemical equilibrium, abundance gradients, and radial

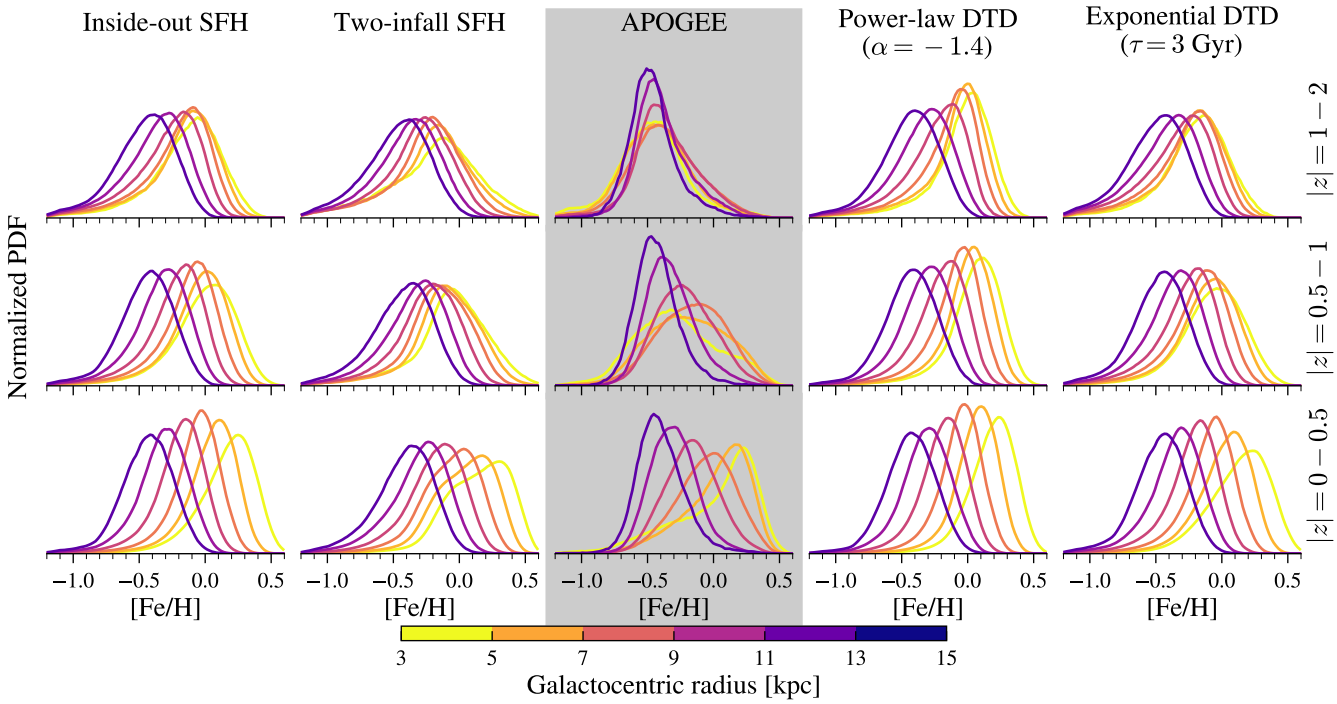

Figure 5. Distributions of [Fe/H] from multi-zone simulations with various models for the SFH and DTD. Each row presents distributions of stars within a range of absolute midplane distance: $1 \leq |z| < 2$ kpc (*top*), $0.5 \leq |z| < 1$ kpc (*middle*), and $0 \leq |z| < 0.5$ kpc (*bottom*). Within each panel, curves of different color represent the distributions of stars binned by Galactocentric radius $R_{\rm gal}$, from $3 \leq R_{\rm gal} < 5$ kpc (yellow) to $13 \leq R_{\rm gal} < 15$ kpc (blue). Each distribution is normalized so the area under the curve is 1, and the vertical scale is consistent across each row. All distributions are convolved with observational uncertainties in APOGEE DR17 (see Table 4) and smoothed with a box-car width of 0.2 dex. *Left columns:* comparison between the inside-out and two-infall SFH models. In both cases an exponential DTD with timescale $\tau = 1.5$ Gyr is assumed. *Center column:* the distributions from APOGEE DR17 for reference, binned and smoothed similarly. *Right columns:* comparison between the power-law ($\alpha = -1.4$) and exponential ($\tau = 3$ Gyr) DTD models. In both cases the inside-out SFH is assumed.

migration (J21, see their Section 3.2 for further discussion).

4.2. *The distribution of [O/Fe]*

The distribution of [O/Fe] serves as a record of the relative rates of SNe Ia and CCSNe. As such, the shape of the distribution is greatly affected by both the SFH and DTD. Figure 6 shows the distribution of [O/Fe] across the disk for the four model SFHs, all assuming an exponential DTD with $\tau = 1.5$ Gyr, compared to the distributions measured by APOGEE. We see similar trends with Galactic region across all four simulations. Near the midplane, the distributions at the innermost and outermost annuli are similar, but away from the midplane, there is a clear trend of a higher average stellar [O/Fe] at small $R_{\rm gal}$ and lower [O/Fe] at large $R_{\rm gal}$.

While trends with $R_{\rm gal}$ and $|z|$ are similar, the shape of the distribution varies greatly with the chosen SFH. The inside-out and late-burst models produce similar distributions because of the similarity of their underlying SFHs —both skew heavily toward near-Solar [O/Fe], although the late-burst model produces a slightly broader peak and a less-prominent high-[O/Fe] tail. The only region which shows any significant skew toward high [O/Fe] is $R_{\rm gal} = 3 - 5$ kpc and $|z| = 1 - 2$ kpc, but the shift to higher [O/Fe] at high latitudes is gradual and does not produce the notable trough at [O/Fe] ≈ 0.2 which is seen in the APOGEE data.

On the other hand, the early-burst model produces a bimodal distribution of [O/Fe] even out to large radii and close to the midplane, whereas the APOGEE distribution does not show a prominent high-α peak beyond $R_{\rm Gal} \sim 11$ kpc. The early-burst SFH produces the closest match to the data by far: the low-α sequence away from the midplane shows increasing contribution with larger radii, with the innermost radii containing very few low-α stars. However, the high-α sequence contains substantial contribution from stars at large radii, which is not seen in the data.

The two-infall SFH model produces *three* distinct modes at [O/Fe] ≈ 0.0, 0.2, and 0.4. At small $R_{\rm gal}$ and with increasing $|z|$, the low-α peak decreases in promi-

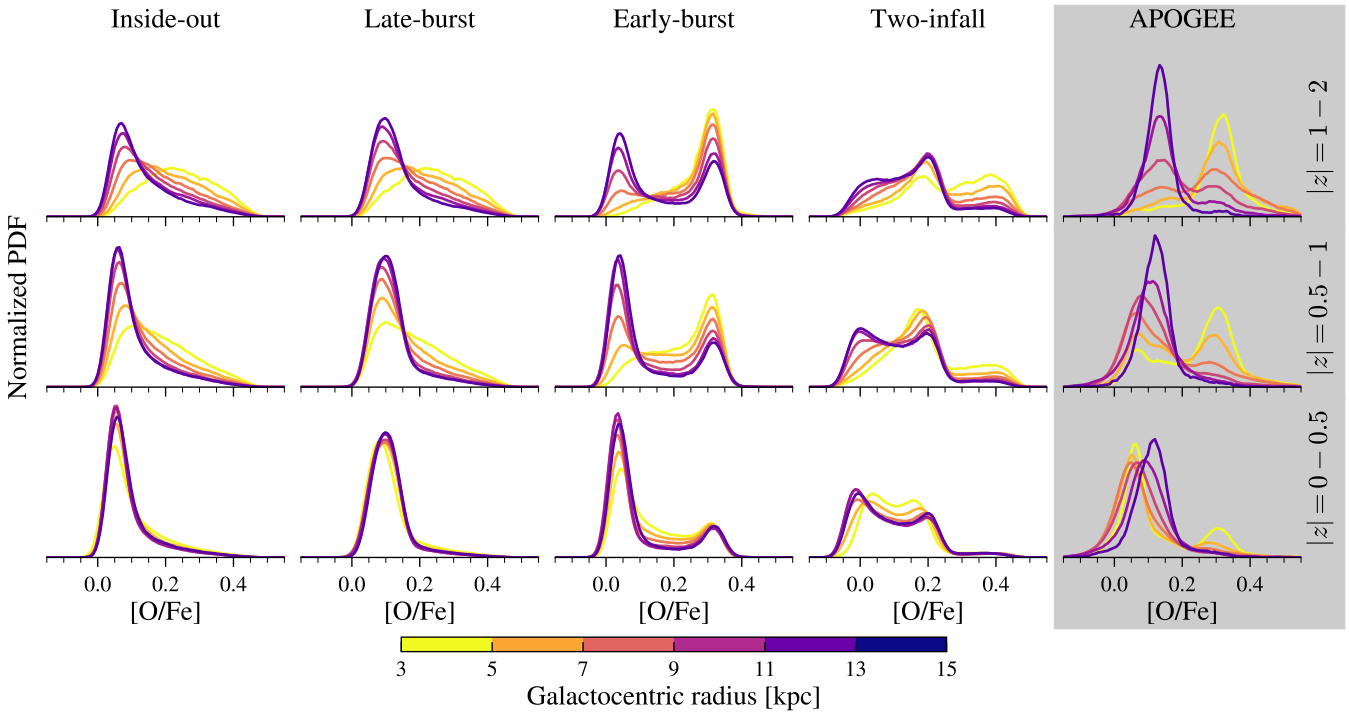

Figure 6. Distributions of [O/Fe] from multi-zone simulations with different SFHs. In all cases an exponential DTD with timescale $\tau = 1.5$ Gyr is assumed. The format of each panel is the same as in Figure 5, except that all distributions are smoothed with a box-car width of 0.05 dex. Distributions from APOGEE DR17, binned and smoothed similarly, are presented in the right-most column for reference.

nence and the high-α peak increases, but the intermediate peak is a striking feature at all latitudes which is at odds with the APOGEE data.

Figure 7 shows [O/Fe] distributions produced by simulations with the same SFH but a range of different DTDs. We choose the early-burst SFH because it produces distinct high- and low-α sequences, which makes the effect of the DTD on the former more apparent. The most apparent effect of the DTD is to shift the mode of the high-α sequence. The two-population DTD, which has the most prompt SNe Ia of any model in this paper, places the high-α sequence at [O/Fe] ≈ 0.2, while the triple-system DTD, which has the fewest prompt SNe Ia, places it ~ 0.2 dex higher at [O/Fe] ≈ 0.4. This metric alone would seem to favor the exponential ($\tau = 1.5$ Gyr) or plateau ($W = 0.3$ Gyr; not shown) models; however,

4.3. *Bimodality in [O/Fe]*

A striking feature of the [O/Fe] distribution in the Milky Way is the distinct separation into two components, the high- and low-α sequences. The relative prominence of the two peaks varies with $R_{\rm gal}$ and $|z|$, as shown by the right-hand panels of Figures 6 and 7. A crucial feature of this bimodality which is not apparent in the previous figures is the coincidence of both sequences at the same [Fe/H]. Figure 8 compares the [O/Fe] distributions in the Solar annulus ($7 \leq R_{\rm gal} < 9$ kpc and $0 \leq |z| < 2$ kpc) in two bins of [Fe/H] for select model outputs and APOGEE data. The APOGEE distributions in the bottom-right panel show that the high-α mode is more prominent at lower [Fe/H] and vice versa, but the distributions in both bins are clearly bimodal. Moreover, the "trough" at [O/Fe] ≈ 0.2 is consistent in each bin.

The top row of panels in Figure 8 shows the [O/Fe] bimodality (or lack thereof) across five different DTD models, all of which assume the late-burst SFH. To quantify the bimodality, we use the `scipy`'s (Virtanen et al. 2020) peak-finding algorithm with an arbitrary prominence threshold of 0.1. Five of the eight DTD models (four of the five shown, plus the exponential ($\tau = 3$ Gyr) model) exceed this threshold, while both power-law DTDs and the plateau ($W = 0.3$ Gyr) model do not.

[Reference Vincenzo et al. (2021) for how the bimodality is made stronger by the selection function, but it's not just due to selection effects.]

4.4. *The [O/Fe]–[Fe/H] Plane*

[Subsection introduction] Section 3 and specifically Figures 3 and 4 illustrated the effect of the DTD mod-

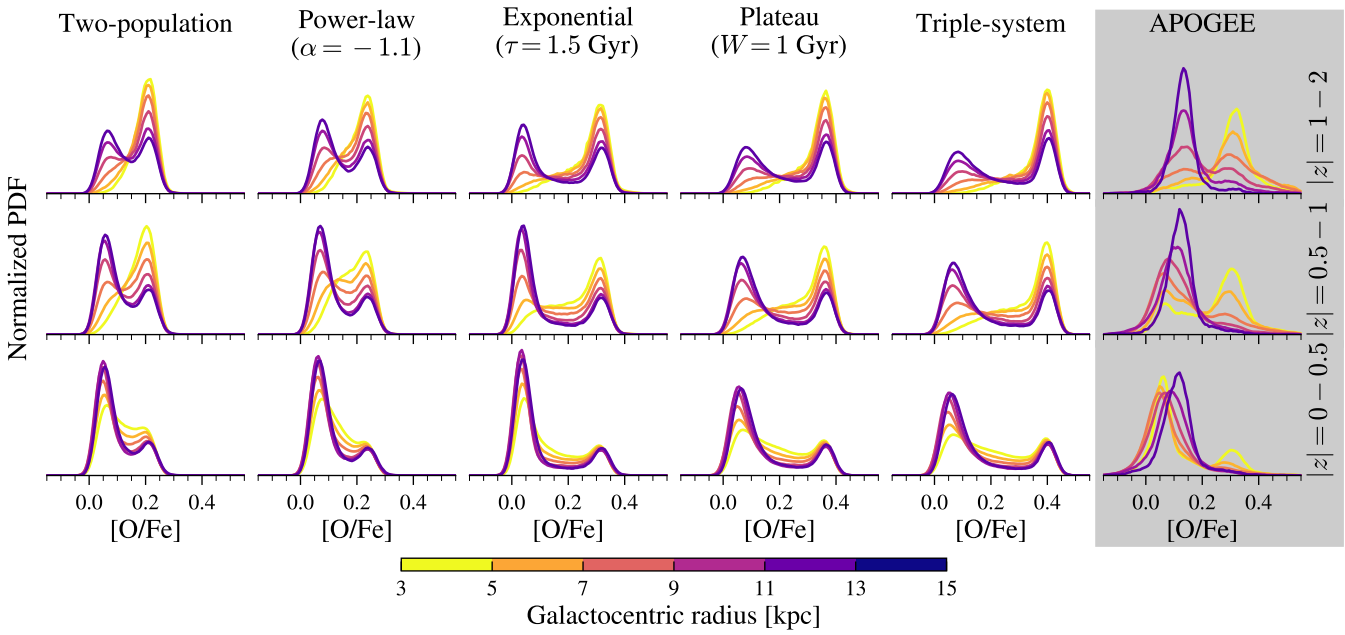

Figure 7. Distributions of [O/Fe] from multi-zone simulations with different DTDs. In all cases an early-burst SFH is assumed. The plot format is similar to Figure 6.

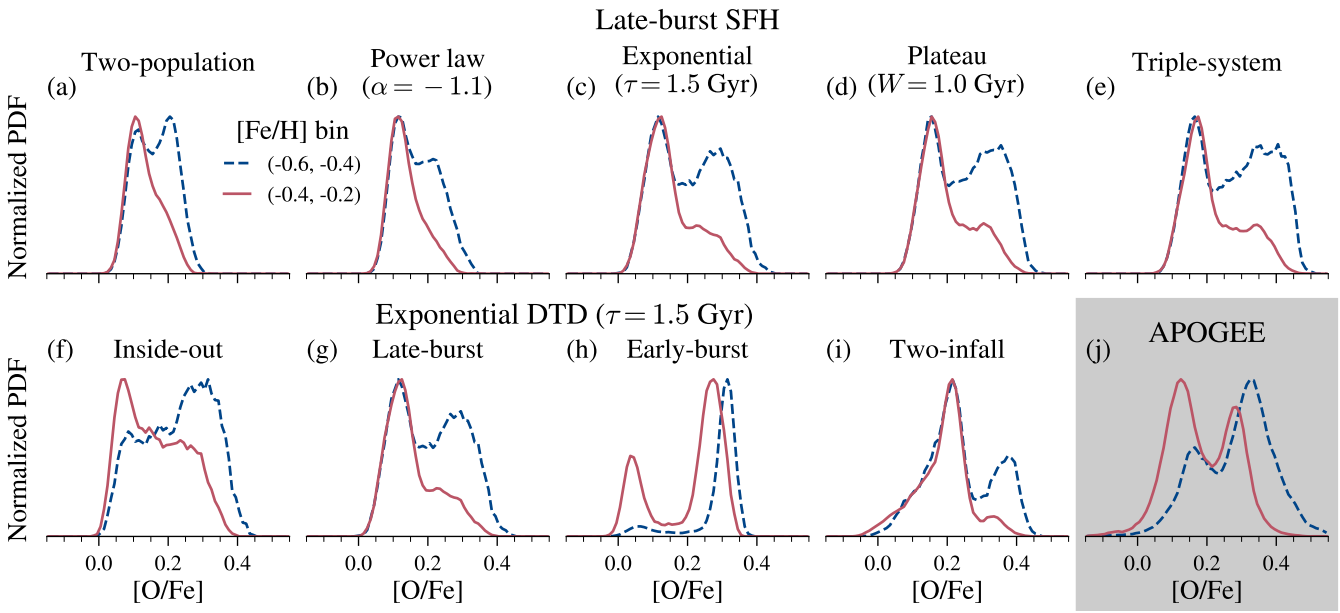

Figure 8. The distributions of [O/Fe] along two slices of [Fe/H]: $-0.6 \leq$[Fe/H]< -0.4 (blue dashed) and $-0.4 \leq$[Fe/H]< -0.2 (red solid). *Top row:* results from five multi-zone simulations which assume the late-bust SFH but different DTD models. *Bottom row*: the first four panels compare the four SFH models, all assuming an exponential DTD with $\tau = 1.5$ Gyr. The bottom-right panel (highlighted) plots data from APOGEE DR17 for reference. All panels contain stars within the Galactic region defined by $7 \leq R_{\rm gal} < 9$ kpc and $0 \leq |z| < 2$ kpc. The $|z|$-distributions of the model stellar populations are re-sampled to match the vertical distribution of the APOGEE sample, and 20 000 stellar populations are drawn from the adjusted distribution in each panel. The maximum of each distribution is normalized to 1 and the vertical scale is consistent across all panels.

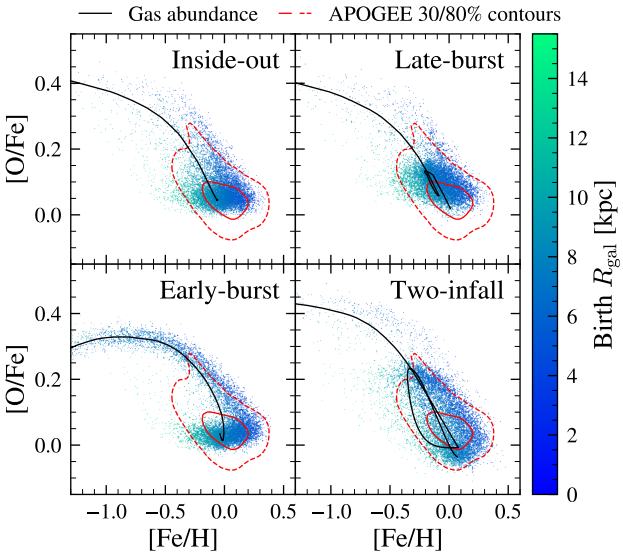

Figure 9. A comparison of the [O/Fe]–[Fe/H] plane between the four SFH models in our multi-zone simulations. All assume an exponential DTD with $\tau = 1.5$ Gyr. Each panel plots a random mass-weighted sample of 10,000 star particles in the Solar neighborhood ($7 \leq R_{\rm gal} < 9$ kpc, $0 \leq |z| < 0.5$ kpc) color-coded by the Galactocentric radius at birth. A Gaussian scatter has been applied to all points based on the median abundance errors in APOGEE DR17 (ses Table 4). The black curves represent the ISM abundance tracks in the 8 kpc zone. The red contours represent a 2-D kernel density estimate with a bandwidth of 0.03 of the APOGEE abundance distribution in that Galactic region, with the solid and dashed contours enclosing 30% and 80% of stars in the sample, respectively.

els and parameters on [O/Fe]–[Fe/H] tracks in idealized one-zone models.

Figure 9 compares the [O/Fe]–[Fe/H] plane in the Solar neighborhood ($7 \leq R_{\rm gal} < 9$ kpc, $0 \leq |z| < 0.5$ kpc) between our four model SFHs. All simulations presented in this figure assume an exponential DTD model with $\tau = 1.5$ Gyr, as it lies in the middle in terms of balance between prompt and delayed SNe Ia. The black curves represent the ISM abundance as a function of time in the central zone; in the absence of radial migration, all simulated stellar populations would lie along these lines. Stars to the left of the abundance tracks were born in the outer disk, while stars to the right of the tracks were born in the inner disk, as evidenced by the color-coding in the figure.

The inside-out and late-burst models produce similar results, the main difference being a slightly broader core of the abundance distribution around [O/Fe] ≈ 0.05 dex. The Gaussian burst in SFH in the late-burst model introduces a loop in the ISM abundance track, where an uptick in star formation at $t \approx 11$ Gyr raises the CCSN rate, leading to a slight increase in [O/Fe] before the subsequent increase in the SN Ia rate brings it back down. This same pattern is seen much more strongly in the abundance tracks for the two-infall model: here, the significant infall of pristine gas at $t = 4$ Gyr leads to rapid dilution of the ISM, followed by a large burst in the SFR which raises [O/Fe] by ~ 0.2 dex. In this case the second infall produces a "ridge line" in the abundance distribution at [O/Fe] ≈ 0.2. Finally, the early-burst model shows the clearest separation between a high-[O/Fe] track which follows the ISM abundance from low metallicities, and a low-[O/Fe] sequence composed of stars with a wide range of birth radii. The early-burst model sees a higher density of points which closely follow the ISM track at [Fe/H] $\lesssim 0.5$ which is likely the result of an increased SFR at early times relative to the other models.

Figure 10 compares [O/Fe]–[Fe/H] distributions for five simulations with the same SFH (inside-out) but different DTD models. We choose the inside-out SFH model for this figure because it produces a monotonically-decreasing abundance track, making comparisons between the different DTD models relatively straightforward. We note that in this model, both the low- and high-α sequences are composed of stars with a wide range of birth $R_{\rm gal}$. The simulation results are arranged by increasing median delay time from left to right. Results for the two-population and power-law ($\alpha = -1.1$) DTD models, with a large fraction of prompt SNe Ia, are presented in the first and second columns, respectively. For these models, the simulated stellar abundances are reasonably well-aligned with the APOGEE contours at low $|z|$ (bottom panels), but they entirely miss the observed high-α sequence at high $|z|$ (top panels). In the center column, the exponential ($\tau = 1.5$ Gyr) DTD produces a distribution which aligns quite well with the APOGEE contours in all $|z|$-bins, and even produces a "ridge" which extends to large [O/Fe] at low- and mid-latitudes (bottom and center panels, respectively). While it does better at populating the high-α sequence than the previous DTD models, the bulk of the simulated populations at large $|z|$ still fall below the APOGEE 30% contour (top panel). The two right-hand columns present simulation results for the plateau ($W = 1$ Gyr) and triple-system DTD models, which have the longest median delay times. The high-α "knee" occurs at a much higher metallicity in these models and is visible in the gas abundance tracks in the upper-left corner of the panels. At high $|z|$, the simulated abundance align quite well with the APOGEE high-α sequence (top panels), but there is a significant

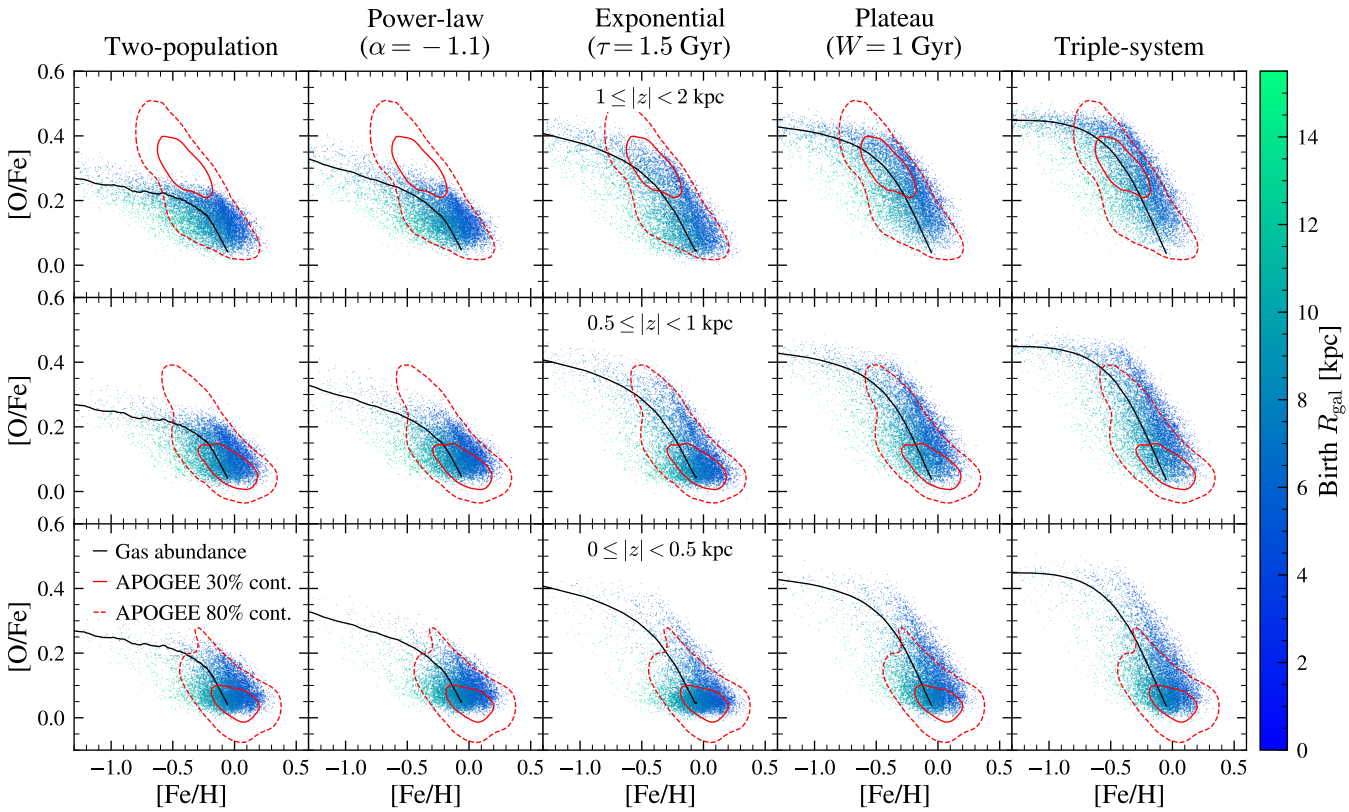

Figure 10. The [O/Fe]–[Fe/H] plane from multi-zone simulations with different DTD models. All assume the inside-out SFH. Each panel is similar to those in Figure 9, except each row contains star particles from a different bin in $|z|$, with stars closest to the midplane in the bottom row and stars farthest from the midplane in the top row. All panels present stars within the Solar annulus ($7 \leq R_{\rm gal} < 9$ kpc).

ridge of high-α stars from the inner Galaxy at low $|z|$ (bottom panels).

[Discuss effect of radial migration]

4.5. *The Age–[O/Fe] Plane*

[Subsection introduction]

Figure 11 compares the stellar age and [O/Fe] distributions in the Solar neighborhood between our four SFH models. Similar to Figure 9, all four panels assume an exponential DTD model with $\tau = 1.5$ Gyr. We include age data from L23 for qualitative comparison, though we caution against drawing strong conclusions because we don't correct for selection effects or systematics in the neural net ages. The inside-out and late-burst models show fair agreement with the data at high [O/Fe], although both show a ~ 2 Gyr offset to older ages. One could shift the ramp-up in star formation to slightly later times or simply run the simulation for a shorter amount of time to close this gap. Like the data, the trend in the median age with decreasing [O/Fe] decreases monotonically in the inside-out model. The late-burst model, however, shows a bump in the distribution around a lookback time of ~ 2 Gyr driven by the Gaussian SFR burst, which is not seen in the data, as noted by J21. Neither model produces a significant amount of stars below Solar [O/Fe], in contrast to the data.

For the early-burst SFH model, the simulated stellar ages are almost perfectly aligned with the data for [O/Fe] $\gtrsim 0.2$. The rapid rise in the SFE at early times delays the descent to lower [O/Fe] values and produces a clump of low-metallicity, high-[O/Fe] stars at an age of ~ 10 Gyr. The early-burst model produces a sharp transition between a linear relationship between [O/Fe] and age for stars older than ~ 5 Gyr and a flat relationship for the younger stars, and it too misses the lowest [O/Fe] bin in the data. Finally, the two-infall SFH model produces the closest overall agreement between the simulated and observed median ages. Stars with [O/Fe] $\gtrsim 0.25$ were produced in the first infall, while the second infall produces a clump of stars with similar metallicity, ages of ~ 8 Gyr, and [O/Fe] ≈ 0.2. There is an interesting small population of very old, low-α stars which is not seen in any of the other models. The two-infall model is the only one which plateaus at sub-Solar [O/Fe] for the youngest stars, leading to a very good match to the observed data at low [O/Fe].

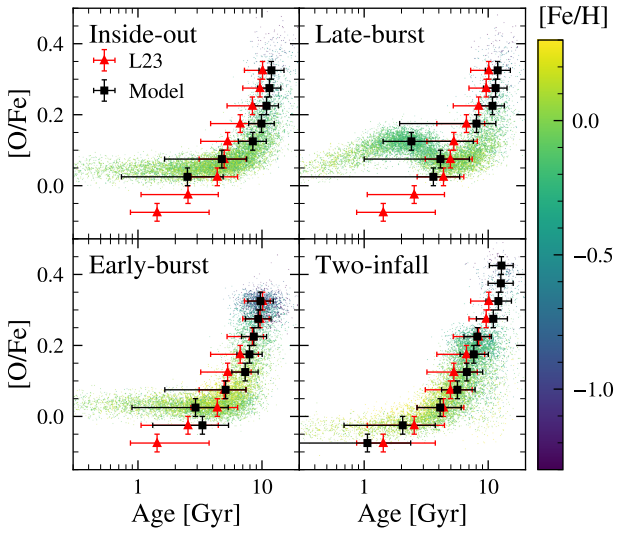

Figure 11. A comparison of the age–[O/Fe] relation between multi-zone simulations with different SFH models. All assume an exponential DTD with $\tau = 1.5$ Gyr. Each panel plots a random mass-weighted sample of 10,000 star particles in the Solar neighborhood ($7 \leq R_{\rm gal} < 9$ kpc, $0 \leq |z| < 0.5$ kpc) color-coded by the Galactocentric radius at birth. A Gaussian scatter has been applied to all points based on the median [O/Fe] error from APOGEE DR17 and the median age error from L23 (see Table 4). Black squares represent the mass-weighted median age of star particles within bins of [O/Fe] with a width of 0.05 dex, and the horizontal black error bars encompass the 16th and 84th percentiles. Red triangles represent the median age from L23 in bins of [O/Fe], and the horizontal red error bars encompass the 16th and 84th percentiles. For clarity, bins which contain less than 1% of the total mass (in the simulations) or total number of stars (in the data) are not plotted.

Figure 12 similarly plots the simulated distributions in age–[O/Fe] space for five of our DTD models. All simulations were run with the early-burst SFH model because the high-α sequence was distinct in metallicity space in Figure 11. Similar to Figure 10, simulations are arranged from left to right by increasing median SN Ia delay time. Moving from left to right, the high-α sequence moves to higher [O/Fe] with increasing median delay time, from ~ 0.4 for the two-population model to ~ 0.4 for the triple-system DTD. As we have seen in previous figures, the range in [O/Fe] produced by the models on the left is much smaller than for the models on the right. At high $|z|$ (top row), the observed range of [O/Fe] is larger than produced by any of our simulations, although the plateau ($W = 1$ Gyr) and triple-system models come close. While the other three fall short of the observed range in [O/Fe], they more closely match the median age–[O/Fe] relation. There is a slight reversal in the observed trend for the stars with the highest [O/Fe]: the $0.45 \leq$ [O/Fe] < 0.5 bin has a slightly *lower* median age than the $0.3 \leq$ [O/Fe] < 0.35 bin at high $|z|$ in the L23 sample, a small effect but one which is not observed in any of our simulation results.

Moving to stars at low $|z|$, our plateau ($W = 1$ Gyr) and triple-system DTDs over-produce stars at the old, high-α end of the distribution, while also diverging significantly from the observed sequence near solar [O/Fe]. The exponential ($\tau = 1.5$ Gyr) DTD comes closest to reproducing the observed range in [O/Fe], while the two DTDs with the shortest median delay time once again produce a smaller range of [O/Fe] than observed. We note that the break between the linear and flat parts of the relation is sharpest for the exponential DTD, and a more gradual transition is observed for the other four DTD models. This could be because the exponential DTD is most dominant at intermediate delay times ($\tau \sim 1 - 3$ Gyr) but falls off much faster than the other models at long delay times, so the evolution of [O/Fe] is quite slow for lookback times $\lesssim 5$ Gyr. Overall, the exponential ($\tau = 1.5$ Gyr) DTD most closely matches the data for stars with $0 \leq |z| < 1$ kpc.

5. DISCUSSION

Extra-galactic observations of the DTD are more sensitive to the late-time DTD; GCE comparisons with the high-alpha sequence are more sensitive to the early-time DTD. [Later on, make suggestions]

5.1. *Scoring the Model Outputs*

The results presented in Section 4 represent a fraction of the total model outputs. We ran models for four different SFHs and eight different DTDs, for a combined total of 32 model runs. While we focused our attention on results in the Solar annulus (for which we still have the best data), we have model outputs which span the entire disk out to 20 kpc. In order to identify trends among our many models, it will be useful to have some semi-quantitative way to compare their outputs to APOGEE data across the disk.

We compare our multi-zone outputs to APOGEE across five diagnostics: the [Fe/H] DF, [O/Fe] DF, [Fe/H]–[O/Fe] plane, age–[O/Fe] plane, and [O/Fe] bimodality. We perform statistical tests between APOGEE and simulation outputs in each region of the galaxy, then average the scores together (weighted by the number of APOGEE targets in each galactic region) to obtain a single numerical score for each simulation. Note that we are not trying to use these scores to fit our models to the data, nor are we trying to rule out any SFH in general. We use these scores to get a handle on which combinations of SFH and DTD are favorable or disfavorable in certain regimes.

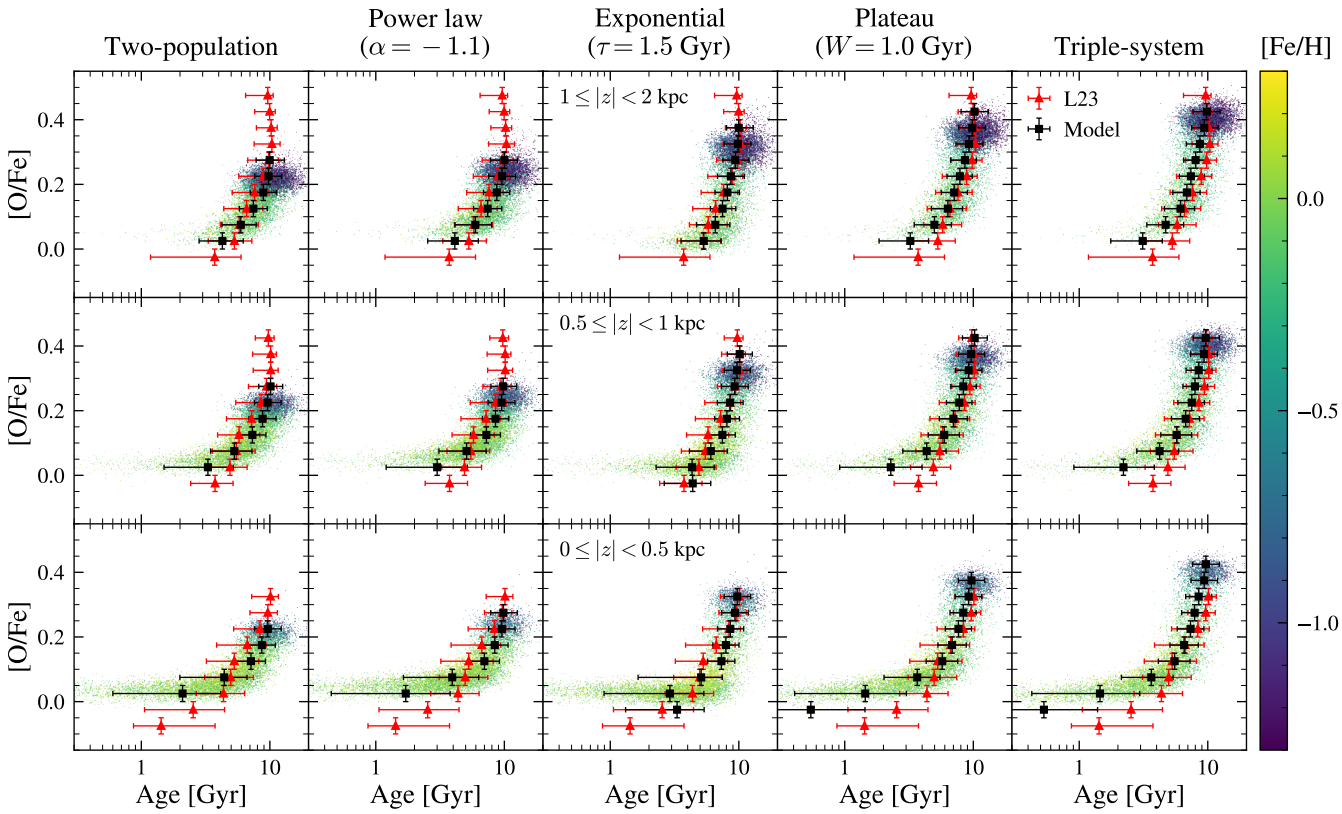

Figure 12. A comparison of the age–[O/Fe] relation between multi-zone simulations with different DTD models. All assume the early-burst SFH. Each row contains star particles from a different bin in $|z|$, with stars closest to the midplane in the bottom row and stars farthest from the midplane in the top row. In all panels stars are limited to the Solar annulus ($7 \leq R_{\rm gal} < 9$ kpc), and the layout of each panel is as in Figure 11.

To quantify the difference between the abundance distributions (MDFs and [O/Fe] DFs) generated by VICE and those observed in APOGEE, we compute the Kullback-Leibler (KL) divergence (Kullback & Leibler 1951), defined as

$$D_{\rm KL}(P \parallel Q) \equiv \int_{-\infty}^{\infty} p(x) \log\left(\frac{p(x)}{q(x)}\right) dx \qquad (18)$$

for distributions P and Q with probability density functions $p(x)$ and $q(x)$. A KL divergence of 0 indicates that the two distributions contain equal amounts of information. In this case, P is the APOGEE DF, Q is the model DF, and x is either [Fe/H] or [O/Fe]. For each model SFH and DTD, we compute $D_{\rm KL}$ in 18 different Galactocentric regions: 2 kpc-wide bins in $R_{\rm gal}$ between 3 and 15 kpc, and bins in $|z|$ of $0-0.5$ kpc, $0.5-1$ kpc, and $1-2$ kpc. The KL divergence for the entire model is taken to be the average of $D_{\rm KL}$ for each region weighted by the number of APOGEE stars in that region.

Perez-Cruz (2008) present a method for estimating the KL divergence between two continuous, multivariate samples using a k-nearest neighbor estimate. We implement this method to calculate the divergence between the multi-zone models and APOGEE data in [Fe/H]—[O/Fe] space.[2]

We employ a different test in the age—[O/Fe] plane. [motivated by the large age uncertainties?] As shown in Figures 11 and 12, in each Galactic region we bin the model outputs and data into bins of [O/Fe] with a width of 0.05 dex. We define the RMS median age difference for the region as

$$\Delta\tau_{\rm RMS} \equiv \sqrt{\frac{\sum_k \Delta\tau_k^2 n_{{\rm L23},k}}{n_{\rm L23,tot}}} \qquad (19)$$

where $\Delta\tau_k = {\rm med}(\tau_{\rm VICE}) - {\rm med}(\tau_{\rm L23})$ is the difference between the mass-weighted median star particle age in VICE and the median stellar age from L23 in bin k, $n_{{\rm L23},k}$ is the number of stars from the L23 age sample in bin k, and $n_{\rm L23,tot}$ is the total number of stars in the sample in that Galactic region. As before, the score for the model as a whole is the average of $\Delta\tau_{\rm RMS}$ across

[2] We got the implementation from https://mail.python.org/pipermail/scipy-user/2011-May/029521.html.

all regions, weighted by the number of stars in the age sample in each region.

The results of these tests are presented in Table 6. Instead of reporting the numerical scores for each simulation output, in the column corresponding to a particular diagnostic we simply write ✓, ∼, or ×, which corresponds to a score in the top, middle, or bottom tirtile out of all of our simulations, respectively. A ✓indicates that particular combination of model SFH and DTD performed better than most when compared to the APOGEE sample for that diagnostic.

[Why does two-infall do so well across the board, even in the [Fe/H] and [O/Fe] DFs where it seems worse by eye?]

5.2. *The Two-Infall SFH Model*

[Compare to previous studies] Discuss Matteucci et al. (2006) findings of the Mannucci et al. (2006) DTD in the two-infall model.

Poulhazan et al. (2018) found that DTD models with a significant prompt component (power-law and two-population) produced the highest average stellar abundance of [O/Fe]. They also found that these models produced the narrowest [O/Fe] distributions and suppressed the long tail to low [O/Fe] ratios. Conversely, their models with essentially 0 prompt SNe Ia produced a lower average stellar [O/Fe] ratio at late times and a broader distribution of [O/Fe]. Their isolated galaxy simulations have an SFR which features an early (< 1 Gyr) burst followed by a roughly exponential decline, qualitatively similar to our early-burst model but with the peak of star formation occuring ~ 3 Gyr earlier. We note that their simulations were not designed to reproduce a Milky Way-like galaxy; as such, their [O/Fe] distributions peak around ~ 0.45 which is higher than the APOGEE high-α sequence. Nevertheless, our study reproduces their finding that a DTD with more prompt SNe Ia will produce a narrower distribution of [O/Fe]. One exception to this is the exponential DTD, which has a marginally narrower distribution than the power-law DTD close to the midplane [numbers here] but a more prominent hith-α tail. This is likely a consequence of the fact that the exponential DTD has a much lower SN Ia rate at late times (~ 10 Gyr) than other models [expand on this].

Palicio et al. (2023) compare seven different DTDs, including the analytical SD, DD-WIDE, and DD-CLOSE, as well as the empirical two-population and t^{-1} power-law DTDs, in a one-zone model with a two-infall SFH. [Note that they use some different parameters for the analytical models than we do). Critically for our purposes, and in contrast to previous studies of the two-infall model (e.g., Chiappini et al. 1997; Spitoni et al. 2021, others), the authors include the effects of outflows, although they explore mass-loading values of $\eta = 0.2 - 0.8$ which is significantly lower than in our simulations.

[Metallicity dependent DTD]

[Is the MW DTD the same as elsewhere?] Walcher et al. (2016) argue that the similarity of the age–[α/Fe] relation between Solar neighborhood stars and nearby elliptical galaxies is evidence for a universal DTD; between a bimodal, narrow Gaussian, and power-law DTD, they find the latter best explains their data.

5.3. *Radial Migration & Bimodality*

One curious chemical feature of the Milky Way disk is the existence of two populations separated by their [α/Fe] abundance ratios at similar metallicity (e.g., Bensby et al. 2014).

The bimodal distribution of [α/Fe] in stars with similar metallicities was first noted in the Solar neighborhood (e.g., Fuhrmann 1998), later extended to the whole of the Milky Way disk (Nidever et al. 2014; Hayden et al. 2015). The "high-α" population contains stars with low enrichment from SNe Ia, so their abundance ratios reflect that of almost pure CCSN enrichment; it is associated with the kinematically hot thick disk (Bensby et al. 2003). The "low-α" population is associated with the thin disk and contains stars with roughly solar α-abundances. Various explanations of the [α/Fe] bimodality have been proposed, including radial stellar mixing (Schönrich & Binney 2009) and a bursty gas infall history (e.g., Spitoni et al. 2021). What separates the two sequences is the ratio of CCSN to SN Ia enrichment, so the DTD ought to have some impact on the [α/Fe] bimodality, perhaps even determining whether or not it exists.

A complicating factor in observing and modeling the effects of the DTD is the migration of stars throughout the disk. Sellwood & Binney (2002) first demonstrated that a transient spiral perturbation (or other potential feature; see [citations]) can cause resonant interactions which adjust stars' guiding center radii without leading to kinematic heating of the disk. Core-collapse SNe explode on timescales of a few Myr, before the progenitor has a chance to migrate far from its birth location. On the other hand, SNe Ia have a broad distribution of delay times and their progenitors can migrate great distances between birth and explosion. The consequence is that SNe Ia can enrich regions of the Galaxy far removed from regions which have experienced a high rate of star formation. Talk about Scannapieco et al. (2005). The first

Table 6. Summary of qualitative comparisons between the simulation output, APOGEE DR17 abundances, and L23 ages for each multi-zone simulation.

DTD	SFH	[Fe/H] DF	[O/Fe] DF	[Fe/H]–[O/Fe]	Age–[O/Fe]	Bimodality
Power-law	Inside-out	×	∼	∼	×	×
($\alpha = -1.1$)	Late-burst	×	×	×	×	×
	Early-burst	∼	×	×	∼	✓
	Two-infall	✓	✓	✓	∼	×
Power-law	Inside-out	×	×	×	×	×
($\alpha = -1.4$)	Late-burst	×	×	×	×	×
	Early-burst	✓	×	×	∼	✓
	Two-infall	✓	∼	∼	×	×
Exponential	Inside-out	∼	✓	✓	∼	✓
($\tau = 1.5$ Gyr)	Late-burst	∼	∼	✓	✓	✓
	Early-burst	✓	∼	∼	✓	✓
	Two-infall	✓	✓	✓	✓	✓
Exponential	Inside-out	∼	✓	✓	∼	✓
($\tau = 3.0$ Gyr)	Late-burst	×	×	∼	∼	✓
	Early-burst	∼	∼	∼	✓	✓
	Two-infall	✓	✓	✓	✓	✓
Plateau	Inside-out	×	∼	∼	×	×
($W = 0.3$ Gyr)	Late-burst	×	×	×	×	×
	Early-burst	∼	∼	∼	✓	✓
	Two-infall	✓	✓	✓	✓	✓
Plateau	Inside-out	∼	✓	✓	∼	×
($W = 1.0$ Gyr)	Late-burst	×	×	×	×	✓
	Early-burst	∼	∼	∼	✓	✓
	Two-infall	✓	✓	✓	✓	✓
Two-population	Inside-out	×	∼	×	×	×
($t_{\rm max} = 0.05$ Gyr)	Late-burst	×	×	×	×	✓
	Early-burst	✓	×	×	∼	✓
	Two-infall	✓	✓	∼	∼	×
Triple-system	Inside-out	∼	✓	✓	∼	×
($t_{\rm rise} = 0.5$ Gyr)	Late-burst	×	×	×	×	✓
	Early-burst	∼	∼	∼	✓	✓
	Two-infall	✓	✓	✓	✓	✓

GCE model to incorporate radial mixing was developed by Schönrich & Binney (2009), and subsequent studies have incorporated more comprehensive numerical treatments of radial migration and the effect of a central bar (e.g., Minchev et al. 2013). In this paper, we follow the treatment by J21, who [describe treatment].

Karapetyan (2022) find a correlation between the SN Ia light curve decline rate, Δm_{15}, and the distance between the SN and the star-forming shock front of spiral arms. As this distance is expected to be an indicator of the age of the SN progenitor, such a correlation would suggest that the less-luminous, faster-declining SNe Ia come from an older progenitor population, which would be expected from progenitor models with sub-Chandrasekhar mass explosions. A similar study on the *radial* migration distance is probably impossible due to the challenges in estimating birth radii even in the Milky Way. It could be interesting to incorporate a SN Ia yield which varies with progenitor age into a GCE model. [Are there analytic DTDs for a DD progenitor with sub-Chandra explosions?]

Something about J21 and Schönrich & Binney (2009).

We need to constrain DTD or galactic SFH better. With SDSS-V, the latter seems much more achievable.

6. CONCLUSIONS

- Our diagnostics consistently favor a DTD with a large fraction of delayed SNe Ia, such as the 3 Gyr exponential, 1 Gyr plateau, or triple system evolution models, over a DTD with a high number of prompt SNe Ia, such as the $t^{-1.4}$ power-law or two-population models. [Discuss the disconnect and/or complement with Maoz & Graur (2017), etc.; something about different timescales probed by volumetric surveys vs. us]
- While the choice of DTD cannot produce a bimodal distribution of [O/Fe], it can affect the shape of the distribution, the ratio of high-α to low-α stars, and the location of the high-α sequence if it exists. The SFH is the critical determining factor in bimodality.
- The distribution of [Fe/H] depends strongly on the chosen SFH, but there is only a weak effect from the DTD.
- The early-burst SFH model is the only one which produced a bimodal distribution of [O/Fe] across the disk resembling APOGEE, from the models we investigate. [Add: summary of why everything else fails.]
- Something about how radial migration can't produce bimodality on its own.

Future work

- Metallicity-dependent DTD normalization and/or shape.

[Personal acknowledgements]

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Software: VICE (Johnson & Weinberg 2020), Astropy (Astropy Collaboration et al. 2013, 2018, 2022), scikit-learn (Pedregosa et al. 2011), SciPy (Virtanen et al. 2020), Matplotlib (Hunter 2007)

APPENDIX

A. REPRODUCIBILITY

This study was carried out using the reproducibility software **show your work!** (Luger et al. 2021), which leverages continuous integration to programmatically download the data from zenodo.org, create the figures, and compile the manuscript. Each figure caption contains two links: one to the dataset stored on zenodo used in the corresponding figure, and the other to the script used to make the figure (at the commit corresponding to the current build of the manuscript). The git repository associated to this study is publicly available at https://github.com/lodubay/galactic-dtd, and the release v.X.X allows anyone to re-build the entire manuscript. The datasets are stored at [URL].

B. ANALYTICAL DTDS

Discuss Greggio (2005) vs our simple function approximations here?

Greggio (2005) derives analytical DTDs for SD and DD progenitor systems from assumptions about binary stellar evolution and outcomes of mass exchange. They find that the parameters which have a large effect on the shape of the DTD are the distribution and range of stellar masses in progenitor systems; the efficiency of accretion in the SD scenario; and the distribution of separations at birth in the DD scenario. The left-hand panel of Figure 13 shows the analytical DTDs for SD progenitors and two different prescriptions for DD progenitors ("WIDE" and "CLOSE"). In the "WIDE" scheme, it is assumed that there is a wide distribution of ratios A/A_0 of the separation of the DD system to the initial separation of the binary, and that the distributions of A and total mass of the system $m_{\rm DD}$ are independent, so one cannot necessarily predict the total merge time of a system based on its initial parameters. In the "CLOSE" scheme, there is assumed to be a narrow distribution of A/A_0 and a correlation between A and $m_{\rm DD}$, so the most massive binaries tend to merge quickly and the least massive merge last.

The right-hand panel of Figure 13 shows the results of one-zone chemical evolution models with the Greggio (2005) DTDs. We assume $\eta = 2.5$, $\tau_* = 2$ Gyr, an inside-out SFH evaluated at a radius of 8 kpc, a continuous recycling approximation, and a minimum SN Ia delay of 40 Myr. We compare the SD, DD WIDE, and DD CLOSE schemes to standard power-law, broken power-law, and exponential DTDs. The SD and DD CLOSE DTDs follow nearly identical tracks in [O/Fe] vs [Fe/H]; however, their distributions on [O/Fe] differ at the low end. The SD DTD follows an exponential with a 1.5 Gyr timescale, whereas the DD CLOSE DTD is well-approximated by a broken power-law with an initial plateau of 300 Myr and a subsequent declining slope of -1.1. The WIDE prescription is likewise best approximated by a broken power-law, but with a longer plateau width of 1 Gyr. In all cases, the difference between the analytical DTD and the nearest broken power-law or exponential is likely too small to be observationally detectable, so in our multi-zone models we implement the more generic functions in lieu of the analytical DTDs.

A multi-zone with the Greggio (2005) SD DTD produced nearly identical results to an exponential DTD with a 1.5 Gyr timescale.

C. STELLAR MIGRATION

For each star particle in VICE, J21 randomly assign an analogue star particle from `h277` and adopt its radial migration distance ΔR and final midplane distance z. This allows VICE to adopt a realistic pattern of radial migration without needing to implement its own hydrodynamical simulation. However, in regions where the number of h277 star particles is relatively low, such as at large $R_{\rm gal}$ and small t, a single h277 star particle can be assigned as an analogue to multiple VICE stellar populations. These populations will have similar formation and migration histories and consequently similar abundances, which produces unphysical "clumps" of stars in the abundance distributions at high latitudes and large radii.

We fit a Gaussian to the distribution of $\Delta R = R_{\rm final} - R_{\rm initial}$ from the `h277` output in bins of $R_{\rm initial}$ and age. Each Gaussian is centered at 0 and we find that the scale $\sigma_{\Delta R}$ is best described by the function

$$\sigma_{\Delta R} = 1.35\,\mathrm{kpc}\left(\frac{R_{\rm form}}{8\,\mathrm{kpc}}\right)^{0.61}\left(\frac{\tau}{1\,\mathrm{Gyr}}\right)^{0.33} \tag{C1}$$

where τ is the age of the star particle.

We fit a sech^2 function (Spitzer 1942) to the distribution of midplane distances z. Vertical migration away from the midplane does not affect the chemical evolution simulation, but we do use z in our analysis. The probability density function (PDF) of z given some scale height h_z is

$$\mathrm{PDF}(z) = \frac{1}{4h_z}\mathrm{sech}^2\left(\frac{z}{2h_z}\right) \tag{C2}$$

and the corresponding cumulative distribution function (CDF) is

$$\mathrm{CDF}(z) = \frac{1}{1+e^{-z/h_z}}. \tag{C3}$$

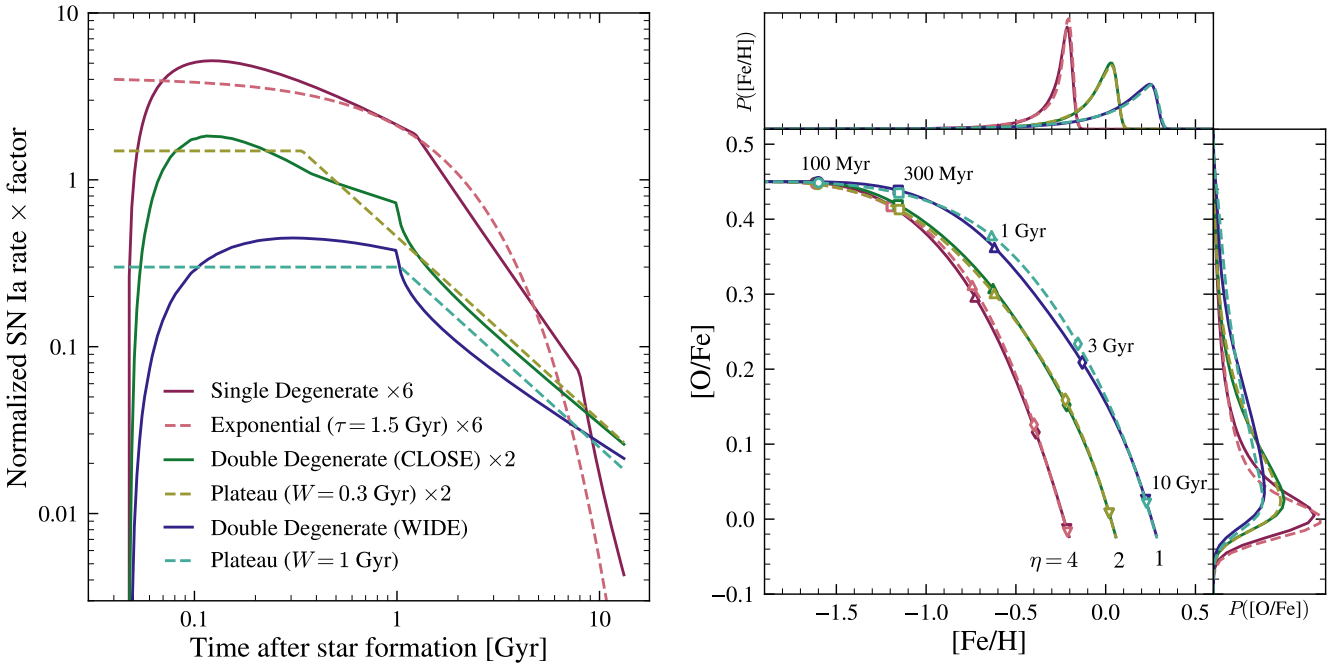

Figure 13. *Left:* Analytical DTDs from Greggio (2005, solid curves) and simple DTD functions (dashed curves). Some functions are presented with a constant multiplicative factor for visual clarity. *Right:* Abundance tracks and distributions from one-zone models with the analytical and simple DTDs (same color scheme). For clarity, we vary the mass-loading factor to be $\eta = 4$, $\eta = 2$, and $\eta = 1$ for the red, green, and blue curves, respectively. All simulation parameters between the similarly-colored solid and dashed curves are identical.

We fit the above function to the distributions of z in h277 in varying bins of τ and $R_{\rm final}$ and found that h_z is best described by the function

$$h_z = (0.25\,{\rm kpc})e^{\frac{\tau - 5\,{\rm Gyr}}{7.0\,{\rm Gyr}}}\, e^{\frac{R_{\rm final} - 8\,{\rm kpc}}{6.0\,{\rm kpc}}}. \tag{C4}$$

When a star particle is created by VICE at initial radius $R_{\rm form}$, we sample its total radial migration distance ΔR from a Gaussian with a width described by Equation C1, and we sample its final midplane distance $z_{\rm final}$ from the distribution described by Equation C2 with a width given by Equation C4. The star particle migrates to its final radius $R_{\rm final}$ in a similar manner to the "diffusion" case from J21, but with a time dependence $\propto \Delta t^{1/3}$. This produces distributions of $R_{\rm final}$ and $z_{\rm final}$ which are similar to the analogue migration case for all but the oldest stars.

Okalidis et al. (2022) studied the stellar migration in the Auriga simulations and found that the migration strength had both a dependence on age and formation radius. They also found that in strongly-barred galaxies, the migration is stronger overall but has a slower timestep evolution than diffusion ($\Delta t^{0.5}$).

Figure 14 compares the distributions of $R_{\rm final}$ in bins of $R_{\rm form}$ and age between the analogue and Gaussian migration schema.

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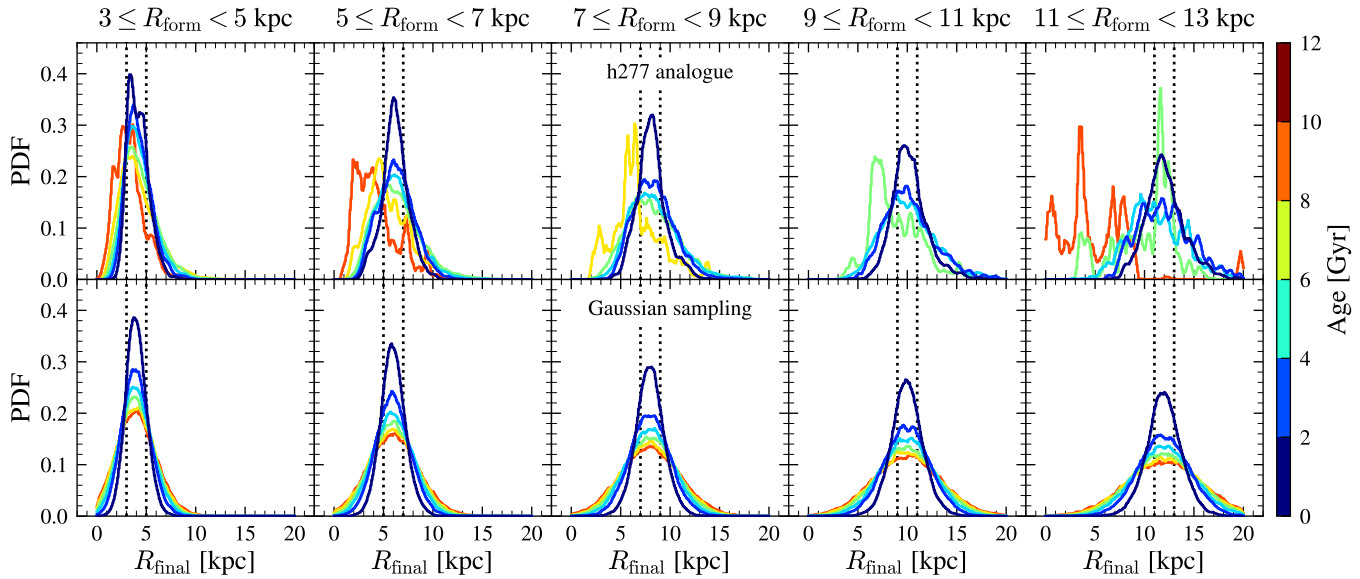

Figure 14. The distribution of final radius $R_{\rm final}$ as a function of formation radius $R_{\rm form}$ and age for the h277 analogue (top row) and Gaussian sampling scheme (bottom row). From left to right, star particles are binned by formation annulus, from the inner disk (far left column) to the Solar annulus (center column) to the outer disk (far right column). Within each panel, colored curves represent the different age bins, ranging from the youngest stars (dark blue) to the oldest (dark red). In the top row, we exclude age bins with fewer than 100 unique analogue IDs for clarity. All distributions are normalized so that the area under the curve is 1, and have been smoothed by a boxcar function with a width of 0.5 kpc. The vertical dotted black lines indicate the bounds of each bin in $R_{\rm form}$; stars within that region of the distribution have not migrated outside their birth annulus over their lifetime.

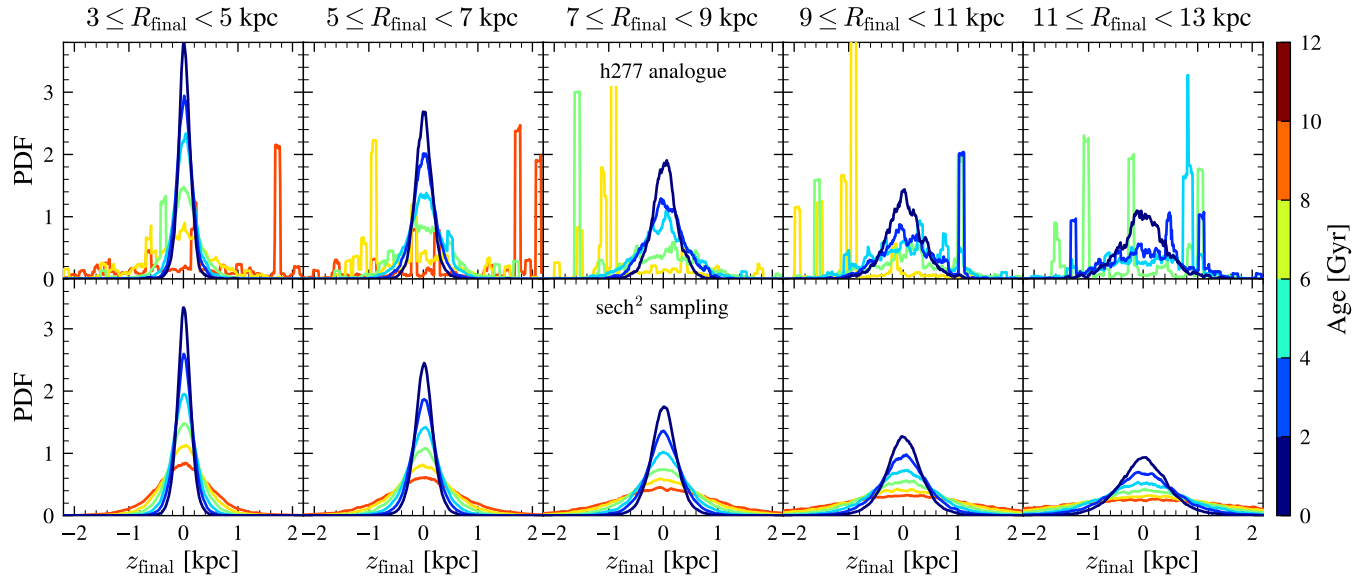

Figure 15. Similar to Figure 14 but for the distribution of final midplane distance $z_{\rm final}$ as a function of final radius and age. From left to right, star particles are binned by *final* annulus. In the top row, we exclude age bins with fewer than 500 unique analogue IDs for clarity. All distributions have been smoothed by a boxcar function with a width of 0.1 kpc.

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